



A survey on internet of energy security: related fields, challenges, threats and emerging technologies

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Received: 26 March 2021 / Revised: 2 August 2021 / Accepted: 19 September 2021 / Published online: 2 November 2021
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Abstract

The need for security has never been more than this for the structure of today's world. Emerging industrial revolutions started to shapeshift many well-known applications to address topics that concern the human race because of the change done to the ecological system regarding carbon emission and environmental pollution. IoE will be the vertical application for this greener innovation projected to have many IoT devices in many fields. The repercussion of mischievous attacks on this new technology would be a calamity in both physical meanings and trustworthiness; the rapid escalation in the number of eco-friendly services developments will have a massive blowback, especially while we are on the 5G doorstep. Focusing on security will help promote the expansion of IoE and greener solutions and the relationship for exploiting the benefits of the smart grid, cloud computing, internet of vehicles. Though this is a new area of research, it has attracted significant attention already. Still, many studies have been done separately to examine the challenges from various angles without reflecting the chronological growth of this technology and security aspects to achieve better strategies. This paper represents a survey related to the surrounding concerns and comparisons for IoE before and after involvement in such services as no survey covers the broad aspect of this topic, discussing its future research opportunities, the raised concern regarding it. Motivated by that to deliver this paper, we identify security as a significant centric topic that can be addressed with innovative approaches and solutions to secure our system through various approaches.

Keywords Confidentiality, Integrity, and Availability (CIA) · Internet of Things (IoT) · Internet of Energy (IoE) · Smart Grids (SGs) · Cyber-Attacks · Internet of Vehicles (IoV) · Cloud-Computing

1 Introduction

Information security is the primary concern of many industries and high importance for many domains such as communications, networking, access management, and software development security. The security pillars are classified into three: (I) confidentiality, (II) integrity, and (III) availability described by the name “CIA triad” as it summarizes all the security goals. Confidentiality is similar

in perspective to privacy, which aims to keep our data safe and allow accessibility only from authorized personnel by protecting the data through means such as data encryption, token, and biometric authentication—integrity refers to the accuracy and trustworthiness of data or protecting the data from unauthorized changes by means like check-sum and server crash. Availability definition is to maintain all the data and make sure that the hardware accessible at any time by authorized persons by adapting ways such as redundancy and backup systems. Security management functions focus on denying unauthorized access by hackers or breaches that can allow data leakage/loss alongside awareness so that the engineers don't miscarry their duties and show that humans can be a significant security factor in the security equation by misunderstanding or not fulfilling a task before or during a specific phase. Now the security direction is trying to elicit accurate security-related requirements during the eliciting stage, which can lead to

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many vulnerabilities such as the implementation stage where the validation of security will lead to insecure applications [1, 2]. IoT is the coordination of multiple machines, devices, and appliances connected to the internet through various networks [3]. According to reports, those interconnected devices, also known as 'things,' will roughly reach around 100 billion by 2025 [4] and such a huge volume of things will make it impossible to ignore the security and the safeness of its application. However, the security side is not well-developed plenty of people ignore that it's not the same as the machine-to-machine (M2M) communication because it's more software-based, not hardware-based. At the moment, we have a different unconventional level of devices and systems communicating using cutting-edge technologies like sensors, as we have them individually or as sensor clusters that demand more privacy, integrity, authentication, anonymity, confidentiality, and availability. The lack of considerations about security issues during the designing and development of these devices generated a big gap making Approximately 70% of the existing IoT devices hold many vulnerabilities [4, 5]. The energy crisis and emission rose universally to a stage that pushed the research to develop a permanent solution for it with a revolutionary vision into the internet by providing a new archetype that needs more scalability, robust infrastructure, and superior communication. However, it will generate an unprecedented volume of data linked and mined to optimize energy efficiency and grow the new users' confidence to shift towards it. Using IoE technology, manufacturers can reduce inadequacies in the current energy infrastructure to increase the generation, transmission, and electricity use to reach its maximum potential. Therefore, this model revolves around a highly complex access environment that is flexible and integrates distributed energy sources that communicate with other power grids while monitoring to assure observability, controllability, and predictability, placing security as the inevitable part internet of energy vision to see its success. Table 1 shows the advantages and concerns of IoE.

1.1 Methodology

This survey represents an investigation for more than 100 research works centered around security. The selected contributions in which authors analyzed and simulated attacks or proposed solutions in a specific area without a bigger picture as the IoE should be implemented and all those aspects work together. We extracted research works that proposed solutions to address various aspects while connecting them to give the reader a bigger picture. Search engines such as Scopus, research-gate, and google scholar were used to identify relevant research alongside prestigious academic databases such as IEEE Xplore, ScienceDirect, and Springer with graduation papers such as thesis and dissertations. The selection factors were recency, citations, and relevance to carry the investigation from various angles. The security pillars (confidentiality, integrity, and authentication also known as CIA triad) were the jugular nerve for this survey to ensure them in any security proposal while adapting new trends to mitigate vulnerabilities. The search criteria included the keywords 'IoE' or 'IoT in the energy sector' filtered by 'surveys or review' to select articles from 2015 until today. The articles were further screened manually by the inclusion criteria that instructed they (1) original work, (2) described actual designs, implementations, and results, (3) addressed the IoT relationship with energy to form the IoE and served IoE use-cases. Figure 1 shows the survey security prospective.

Demonstrating the three essential components for future IoE, which are the smart grid, internet of vehicles, and cloud and how they interact together alongside the obtained benefits and how one of them, once abused, can compromise the others refereeing to recent attacks that have happened with their aftermath and showing the technological advancement in both the academic and industrial sector for dealing with these surfaced attacks. Further, future research directions were disclosed and discussed to provide readers with a multidimensional survey to hope that readers can address IoE security from their aspect of

Table 1 IoE brief description

Categories	IoE	Advantages	Concern
<i>Characteristics</i>	Energy and information exchange	Energy utilization increase stability	Infrastructure security operational security
<i>Service</i>	Combine heterogeneous sources distributed energy sources	Generation steadiness and quality	Cyber attacks hackers
<i>Equipment</i>	Energy routers and smart routers	Increase accuracy and productivity	Reliability data security transmission of energy
<i>Function</i>	Data and energy transmission energy management grid management	Data management energy storage generating power	Lack of standardisation security privacy CIA triad

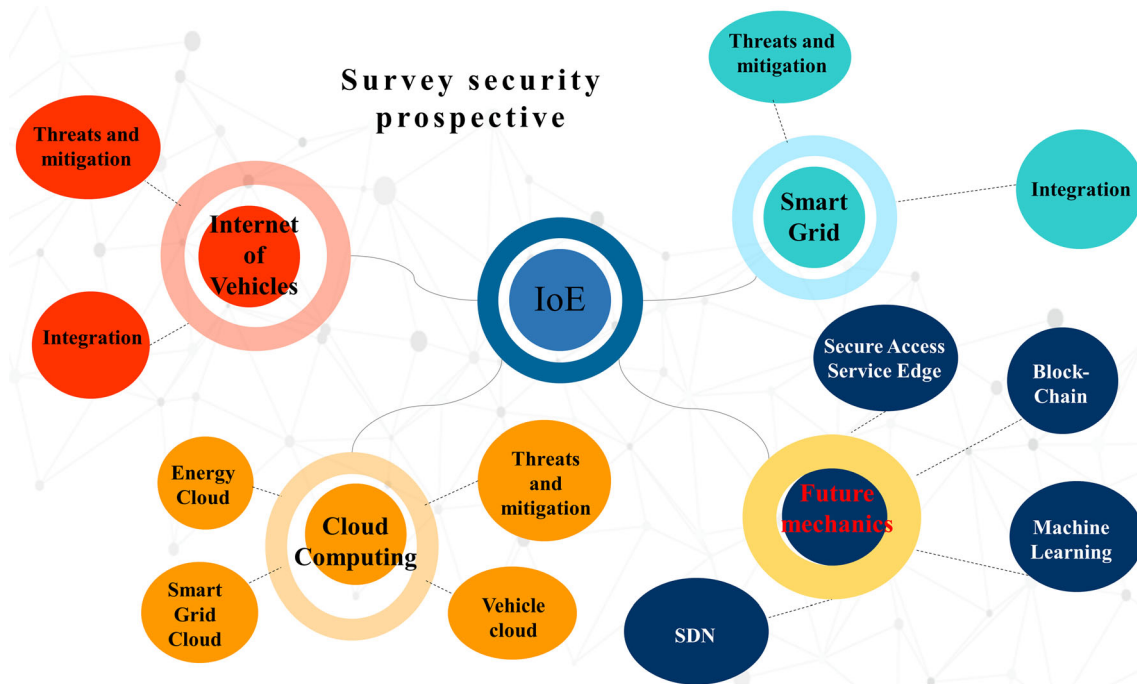


Fig. 1 A map for this survey from a security prospective

interest. Insightful findings, outcomes, new trends, and research problems were present and how a proposed policy can help fill the gap regarding awareness because of the crucial human factor role and economic impact. In aiming to pave the way for future research endeavours tackling the IoE security implementation into actionable information. The main contributions of this paper are summarized as follows:

1. We provide an overview of IoE and its components with a brief introduction to the background, including comparisons and challenges about the IoV and IoE, smart grid, and cloud.
2. We stated the security and privacy requirements and summarised the most prominent security and privacy problems around the CIA.
3. Suggesting a policy that can reduce the knowledge gap as the attacks are unpredictable to happen and the human factor has an important role in IoE
4. We provided classifications of attacks and threats showing their manifestation areas to categorize them based on security objectives and functions. Then, we provided probable solutions
5. Several examples of recent security attacks and their aftermath on the industry or state and we investigated industrial reports to show which solution is more trending.
6. Providing deep insights into future mechanisms and their research directions in this paradigm. Discussing future opportunities for future mechanisms used for

attack mitigations highlighting which issues have been investigated and which methods have been utilized in the IoE and its components.

1.2 Chronological order

To better understand the Internet of Energy (IoE) and the intersections with its components, we look chronologically at the existing surveys on this topic to understand the established needs for our survey from a broader concept and areas on the IoE surveys that considered IoT in the energy section security mainly.

In 2015, the authors [6] surveyed existing protocols for defending communications on the IoT while it's compared against a set of vital security necessities and highlighted the open challenges and strategies for future research work.

In [7], they surveyed the smart sensors networks opportunities for smart grid considering the demand-side energy management and distributed storage by processing and analyzing sensors data. It reveals deeper insights that can help experts improve the power grid's operation to achieve better performance. This manuscript provided recommendations for the future of the smart grid and IoT and the integration of renewable energy sources to attain sustainable energy.

This document [4] is intended to aid the internet society in navigating between dialogues surrounding the IoT about competing for predictions in the promises and perils to pave the way for combining computers, sensors, and

networks for monitoring and controlling devices with critical technologies and market trends.

In 2017, this work [8] considered the future of power systems in the existing smart grids to improve productivity and consistency with renewable energy integration, and they viewed IoE integration through communication networks but only from the smart grid perspective.

Ruland et al. [9] This paper summarizes significant guidelines and standards focusing on smart grids and industrial security while they are compared against applications for smart grids development for future usage.

This survey focuses more on energy-efficient mechanisms to tackle IoT security from the best of knowledge as addressing the security issues and energy impact on energy consumption with what is considered to be the main factors affecting the application of energy-saving techniques for utilizing energy efficiency in IoT security for implying a change in the security protocol, and Implementing the low-power security protocols [10].

In 2018, Kamalrudin et al. [2]. Investigated the demands for IoT applications to secure the information contents that several entities in the applications should deliver while considering the high cost of delivering the services in these applications from the software engineering perspective of the industry scenarios.

In 2019 A survey on the Internet of vehicles: applications, security issues and solutions by Sharma et al. Presented the security requirements and challenges for verifying the protocols for Radio-Frequency Identification Devices (RFID). It has proposed solutions to overcome existing disadvantages in low detection time, low CPU, and memory consumption also compare possible security threats in IoE [11].

Sani et al. [12] Discussed the improvement in IoT deployment into the smart grid infrastructure and illustrates how the smart grid's current structure has become insufficient in meeting the security of energy domains. In addition, proposing a cybersecurity framework that can provide adequate security, privacy, and support energy management in IoE.

In 2020, [13] this manuscript reviews the present works on IoT applications in the energy domain and systems, focusing on the context of smart grids and addressing IoT-enabled technologies, including cloud computing and several platforms for data analysis with the associated challenges in deployment. A systematic study discusses IoT and cloud computing concerned mainly about their combination, which has new challenges and issues because of the rapid integration in many sectors by detecting the challenges and metrics IoT and cloud combination and aims to boost this integration [14]. Butun et al. [15]. This work consists of a detailed review from a security perspective of attacks and challenges towards WSNs in IoT

also discussed the remarkable appealing features of wireless sensor networks (WSNs) such as production cost, autonomy, and longevity for IoT infrastructure along with detection and prevention techniques for security.

This paper [16] represents a survey that contains a compact review of cybersecurity concerns and threats in smart grids regarding the applications and the current and future trends to understand smart grid security further alongside the associated solutions on IoT-based smart grid applications. A survey consists of a three-layer framework addressing the sensing layer for data acquisition, vehicle communication layer, and control layer for automobile functionality in the inter-vehicular internet addressing the threat and attacks in each layer along with countermeasures [17].

One can see that the IoE concept through these surveys is not broad because all these surveys focused on a specific area or two-component integration without the enormous picture impact, scalability, and context within cross-domain. As we have stated in our chronological study of previous surveys and reviews, this exact combination of areas, focusing on IoT in the energy sector and IoE transformation, has not been explored in-depth to the best of our knowledge.

Outline of the Paper: This paper will be alienated into sections related to discussing IoE and the followed methodology, then section two will be for IoE and IoT relationship for the energy. Section three will be for IoE and cloud computing to establish an energy cloud. Next, section four will be for smart grids, section five for the internet of vehicles, section six for the surrounding policy, followed by section seven for threats in IoE and IoE based components and classification of attacks manifestation and impact on IoE. Afterwards, solutions and key technologies can be used as countermeasures, followed by future mechanics we can rely upon for the next generation. Section ten include the future directions. Lastly, some concluding remarks are presented.

2 IoT and IoE

How IoT and IoE Relate The IoT expansion will unleash a new dimension of services for the consumer, providing the potential to deliver solutions that dramatically improve the energy sector efficiency, with other applications that include but not limited to health and education for enterprises to reinforce decision-making schemes and manufacturing productivity. When we take the IoT notions and make it more energy-based and project it, we will have it doing a more comprehensive analysis previously unavailable or has been neglected to most. Likewise, IoE lacks standardization like IoT, but the existing small models

followed the layers principles by dividing the model into layers to process the massive amount of data according to physical specification, functionality, and principles. Namely, perception Layer for data acquisition, network layer that connects the layer as a bridge for communication and exchange, middleware layer that includes computation, data storage, processing, and an application layer that is the management interface for the layers. Some models strived to add more security in devices by following the International Telecommunication Union (ITU) architecture of having a five-layer structural design [3]: the sensing layer, access layer, network layer, middleware layer, and application layer putting all the wired data acquisition devices in the sensing layer and the access layer possesses the wireless devices. The attained IoE will overcome these inefficiencies in the existing energy infrastructure such as SCADA which limits information sharing and exchange, especially in wider numbers of devices in a network-by the growth of the insufficiency corresponding to the number of devices- but with IoE, the scale will grow proportionally with the prospect of generating supplementary values built upon data that is deeply analyzed. Another resemblance between how the energy and the information flow is in routers, whereas IoT and IoE use routers in their applications, as normal internet uses network routers and IoE uses energy routers [18]. In this sense, IoE has been built based on green innovation to create and implement eco-friendlier resolutions and empowering environmental approaches at the consumer and industrial sides for energy well-organized delivery. Green IoE replicates this perception, including counting the collection procedure, practices, manufacturing, disposal means of these resources, habitual methods, and recycling process to extend this humble notion to be used in hardware. Consequently, it will make our system even less hungry for power consumption and shrink the negative impact on nature [19]. IoE is becoming a global groundwork for information enabling services and the best method to integrate different renewable energy resources with old grids to shift towards smarter ones [20]. This awareness perspective in the investigation will facilitate the way to imminent solutions and replacements by taking ecological technologies that will be lessening the carbon emission such as RFID- (Radio Frequency Identification), Cloud Computing, Wireless Sensor Networks (WSN), NAN- (neighbourhood area network). Green IoE will be promoting the energy market and industrial upgrading as its core is creating new energy values [21]. Narrowband IoT (NB-IoT) is a friendly green technology that can fall in the scope of the green IoE by using a narrowband for the process to be described as “greener than any technology [22] that posses a standardized architecture for various working environments that contains six-layers paradigm: the physical layer, MAC layer,

radio link control layer, packet data convergence layer, radio resources control, and non-access stratum. Though many security mechanisms presently exist aiming to enhance security, IoE devices operate in an unattended manner with partial or not existing resistance of tampering to ensure integrity and confidentiality. The poorly designed devices will allow the execution of randoms commands that can cause miscommunication or unauthorized access. It has been estimated that 48% of the users are unaware that the compromised connected devices can be used to launch cyberattacks or to open backdoors to the system, and the biggest portion of users don’t perform firmware updates as they see it as the responsibility of the manufacturers to remediate the risks. Cisco’s surveyed in 2016 [23] and published the results stating that network products such as switches and routers have an average of 28 exposures per device as companies fear reputational damage. The lack of a secure by design approach challenges the IoE growth making it not well-equipped for future endeavors as the designers focus more on functionality than protecting the users. Because data protection is not observable, and the users trust the developers blindly for that reason, having de-facto standards acceptable by both parties of the equation, the public and market forces are necessary [24]. Additionally, the cognitive capability, which can direct a huge volume of data without human intervention and the powering capabilities for these devices and consumer illiteracy, is the hackers’ best friend.

3 Internet of energy and cloud computing

Cloud Computing, also known as a big data center (based on-demand), is a platform acting as a front-end to give more resources or extra memory. It consists of two parts (with direct exchange): the front-end where our client and the back-end where data are stored. By 2020, according to Statista, there will be 30 billion interconnected devices, and by 2025, this number will surge to be 75 billion [25] and all those things will demand tremendous storage. Since IoE about connecting heterogeneous objects and the configurable cloud nature facilitate that [26]. Numerous gains result from the integration between cloud computing and the internet of energy because the capabilities of assisting IoE are boundless with limitless resources combined with tremendous storage capacity. It makes the cloud compensate for some limitations that IoE needs to address as it fits the IoE sensitivity of time, conditions, and pervasive accessibility. The cloud platform possesses and keeps offering several benefits to IoE together that include computation, network bandwidth, and virtualization (connecting numerous users to cloud servers remotely by machines). The cloud uses the shared pools and the

application that depends on it permits the consumers to accomplish computation by spending service resources, delivering the storing entirely over-Internet [27]. Similarly, IoE can support cloud computing by stretching the cloud's scope to add more heterogeneous things to include the natural world in a cost-effective relationship [14]. As shown in Table 2.

Somehow, they are the accompaniments of each other, smoothing what seems like a distinction between them to look as if it's unobserved. However, the connected things generate a possibility of allowing hackers to access them remotely, as recent research showed that the back-end implementations are more vulnerable than devices. Many embedded systems work in constrained situations, e.g., with processing and memory capabilities, which can ultimately reduce the protection mechanisms that can be applied. Additionally, the CIA triad isn't a priority at the designing phase, so new technologies should offer a wider range of protection, specifically integrity and data protection.

What makes the cloud more popular is that public cloud providers have gained a great reputation to address plenty of concerns because the cloud is about back-end implementations and provides quick scalability and resource utilization. The combination of IoE based on the cloud has been introduced in health care [28], smart cities [29], smart logistics [30], video surveillance [31], monitoring [32], and smart energy [33, 34]. Fog computing is on par with cloud computing and exceeds it in many areas; we stated that the cloud is a pool of shared resources between devices connected by the internet. To obtain services is done by using DCs that have remote access or industrialized platforms or else software services. Instead, fog computing came to schedule the tasks among numerous servers. The risen concern regarding the growth of the cloud that it can have downtime, meaning get affected by network issues, and the data are being transmitted globally, to ensure the speed is mainly dependent on the virtual machines, which can be costly. A new concept derived from nature has risen called fog or cloudlet for utilizing private clouds because fog is closer to the earth than clouds. Instead of centralization, we placed the nodes at the edge to be closer to the devices alongside the processing capabilities to be a decentralized

process, so the clouds or nodes don't need consultation every time.

Fog is called the enabler of IoE. After all, in the edge, it needs gateways because it helps only the nodes around it, which adds discretion and then determines what can be processed locally or globally [35] to overcome the latency issue by its topology. To achieve that, it uses friendly communication protocols like Zigbee, Bluetooth which are not only Internet protocols (IPs) to avoid communication failure with the assistance of hubs to process data to bears better heterogeneity towards IoE and location awareness instead of cloud partiality. The exiting shortcomings rest in the scalability, adding more complications to the system and purchasing its own associate devices. Therefore, edge computing came into the picture, similar in concept with a modification in the data processing place, which is done in the devices instead of the edge resulting in enhanced privacy. The device can undergo a pre-processing stage, then analysis followed by transmission to a larger node or gateway. The data can travel from the device to the gateway and then the back-end because it has less access to resources. The process will be done in the back-end or distributed back to the edge. Nevertheless, sacrificing the processing capabilities generated more security because fog is more favored by middleware companies and edge computing by data processing centers.

Another angle to view cloud computing is by private clouds, public clouds, and hybrid clouds. The Public cloud providers attain high credits in ensuring availability and integrity, but not in confidentiality as it depends on topographical locations and compliance with regulations of the providers. On the other hand, private clouds score highly on confidentiality and integrity but not on availability due to latency because of adding new devices and congestions, which can cause data leakage or drops. As a result, the hybrid cloud came to get the better of both worlds dealing with multi-cloud environments by separating both sensitive and critical data in a distinct environment managed by software as a private cloud and benefits from the tools, expertise, and resources that come with the public cloud. The hybrid cloud will orchestrate operations and management to produce a unified distributed computing that integrates cloud resources, monitor, supervise allocation, and

Table 2 Brief comparison between IoE and cloud

IoE	Cloud
The real world connected things	Virtual resources for connected things
Limited computational capabilities [36]	Unlimited computational capabilities
Small or not existing storage capabilities	Humongous storage capabilities
The internet is a way for convergence	Uses internet to deliver services
Need scalability	Quickly scalable
Source of big data	Manage big data

management. In a sense, cloud computing focuses on resources and uses autonomous software to achieve automation inside the cloud to manipulate the resources programmatically from the providers.

A paramount principle used as a protagonist inside the hybrid cloud is Infrastructure as Code (IaC); this automation approach lets developers create flexible environments for changes, written as human-readable languages, and the resources created when executing a code placing extra significance with the arrival of edge computing by moving workloads to actual computation place. Lift and shift is an additional important aspect in hybrid cloud that allow migration of an application from the on-premises to the cloud with software virtualization prospect in hybrid clouds that boost the compatibility with hardware equipment to enrich the end-user experience while reducing the efforts and ensuring the CIA triad without compromising security. In brief, the hybrid cloud will help implement new technologies, elevate the quality of infrastructure, business acceleration, enhance productivity, better compliance, and optimize efficiency [37]. When dealing with Quality of Service (QoS), as it will always be unlikely to guarantee its steadiness, cloud computing introduction in energy internet can allow the QoS to be measured at all the stages of the system scheme in parallel with any cloud-IoE by using multiple cloud service providers as an excellent platform to take charge of the security operation within centers. Also, a separate virtual platform can be used to train the staff in preparing for defending scenarios if an attack happened to increase their skills. Additional benefits for the cloud-IoE is providing scalability to the security level by increasing the security scale according to what seems demanding and granular security, so we can discharge malicious threats before they reach the network. Any update will spread across the network instead of going device per device update.

Cloud computing dares are in technical issues while providing operational and intelligent solutions to meet the resource demand with a supervision mechanism over management as well energy efficiency in the cloud mostly in data centers (DCs) is considered an immense challenge for the reason of the large energy consumption in DCs. The exploitation of embedded systems, RFID, and sensors by software will be extensive [38]. However, accessing many resources without hampering the design and dealing with insufficient bandwidth are challenges that need considerations.

4 Smart grids (SGs)

The phrase grid origin is coming from electrical grids. However, smart grids are an advanced form of grids like an electricity network that allows a bidirectional flow of data and its intelligent enough to route this energy from centralized or distributed plants; this anticipation will make us bid farewell to the fossil fuel-dependent network. The SG involves communication technology, connected power systems, control technology, and intelligent meters to utilize renewable energy sources (RESs). Under the SG umbrella, SG will manage the energy packets just like data packets when they go across routers and gateways access by the notion of a routing algorithm looking to optimize or find the most suitable route to reach the destination. In this perspective, IoE can be seen as a type of infrastructure connecting things and bidding them with protocols to operate and interoperable these communications permitting a real-time balance between the sinks and generation sources [39]. Hence, both energy producers and energy consumers are both well involved. The domains of intelligent grids are [40]:

generation sources, transmission lines, distribution nodes, operation centers, consumption markets, grid customers, and service providers. In a traditional grid, supposing an attacker or a device is compromised in such cases, the whole grid will be exposed because they need physical access to the facilities or inside jobs to tamper with the grid set-up or data leading to catastrophic events that can involve blackouts, service denial, overloads and explosions. Therefore, grid intelligence has a significant role to play inside the grid to attain an efficient energy management process. Unfortunately, it has not been studied in this current work. IoE involvement in smart grids will help efficient utilization of energy usage by controlling it without human intervention. Hence, increasing the effectiveness and increasing the grid-scale. Its disruptive fashion of control will require sensors for remote sensing to boast the grid working capabilities and increase smartness in both manners and its ability to measure, monitor, and communicate. We can monitor the SGs better by smart meters, sensors, and intelligent gadgets to meet the demand response in adding predictive maintenance or consumption optimization [9]; each part can handle information and data provided by its neighbors. Smart meter inside the grid uses Advanced Metering Infrastructure (AMI) to support the automation. It's the zenith of smart meters, networks, and management systems. The meters are for gathering energy usage data periodically, conveying it to utility servers, status upgrades, and the analyzed gathered data exchanged by the embedded communications networks. Eventually,

the management system utilizes the data for billing, monitoring, and troubleshooting.

IoE will reduce resource wastage by optimizing the distributed generation as renewable energy sources face uncertain status regarding the load or the renewable energy sources. Table 3 shows a comparison between the smart grids and IoE smart grids to highlight the gained benefits.

The authors in [41] established smart networked microgrids based on the NSGA-II algorithm, considered as a sorting algorithm to describe the fluctuation inside the grid and pollutants emission. They solved the problem with two heuristic algorithms, the particle swarm optimization (PSO) and Imperialist competitive algorithm (ICA), to minimize the operating power and emissions as the grid service provider will also generate revenues and adequate power supply for users. The NSGA-II algorithm's role is to lessen the overall power usage as it consists of generation power, purchasing power, vented power and utilize the emissions based upon the probabilistic behavior. The optimal scheduling for intelligent homes was studied in [42] by using mixed-integer linear programming (MILP) for minimalizing the predicted energy consumption by scheduling the tasks based on pricing rate and the renewable energy production even in peak demands scenario using 90 home and 30 building alongside their daily life electric appliances and the demands depends on the operation used to find the least cost and power demand. MIPL will schedule the generated power from the renewable source to save power and reduce footprint by this, the load burden on the central grid will be minimal and satisfy the increasing energy demands without altering the living habits drastically, and the author showed that this scheduling changed the people behavior to save more as this model consider the problem from the customer point of view.

Smart grid utilities will replace the outdated grids, but they face a dilemma: networks that link the control stations

with substations and the networks within the sub-stations as the grid possesses advanced peer-to-peer infrastructure. At the same time, the changing supply and demands need more effectual energy delivery with visibility and choices to the user side. Therefore, the security demands can't be an afterthought in the grid but more of a jugular vein with the increasing percentage of successful intrusion or intrusion attempts, attacking the digital meters infrastructure can lead to a breakdown state by allowing surge or over-voltage scenarios setting off the destruction of appliances as well as the meter continuously keep collecting data, it can be used to change its main function, for example, to invade the privacy because the meter will keep recording the behavior of the people and their energy usage patterns [43].

5 Internet of vehicles (IoV)

One of the green innovations for development is making an intelligent platform that combines the automobile industry with the internet allowing urban transportation to meet the information services. IoV will reduce emissions, improves traffic congestion handling, and tackles pollution problems that the new industrial revolution wants to address. IoV will help us step more steadily towards smart cities given a new insight into vehicle interactions with the real world to achieve full supervision with functions such as intelligent vehicular management and intelligent vehicle control [44].

Essential technologies of IoV systems are [45] sensing technology to acquire physical meaning, communication technology, and data processing technology. For an IoV System, communication is the beating heart between the connected vehicles, and it contains five types of communications [46]: Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Roadside Units (V2R),

Table 3 Brief comparison between the smart grid and IoE smart grid

Smart grid	IoE smart grid
Integrate instrument and sensor in electric grid	Integrating advanced instrument and sensors
Don't offer real time monitoring and control	Offer real time monitoring and control
An electric grid with advance communications	All-in-one internet and information sharing
Manual configuration for peripherals and bidirectional flow	Operating system detect and configure peripheral (plug and play)
Requires user intervention	No intervention by users
High computational cost	Less computational cost
Less heterogeneous	More heterogeneous
Traditional operations affected by failures	Failure correction and fault tolerance
Cooperate with electric grids only	Can cooperate with other energy grids

Vehicle-to-Sensors (V2S), and Vehicle-to-Personal Devices (V2T). As shown in Fig. 2.

What's more, the current shape of IoV as stand-alone systems will hinder its growth, deeply affecting its users in terms of energy and time, resulting in energy underutilization with challenges in power degradation. As shown in Table 4.

Similarly, consumers have to wait for a long time to charge the vehicles, realizing some of them can't afford it with concerns about how long this charged battery will last and the distance it will cross. The introduction of IoE with IoV will improve the mobility of these vehicles as they can charge on the go to save time for users and address connection problems that occur because of vehicle movement. Further, enhancing the safety on roads because the vehicle will communicate with the transportation infrastructure using Road Side Units(RSUs) transceiver, limit emissions, provide user safety and minimize hazardous scenarios [46]. In the case of an accident, it can send a notification to ask for medical assistance or showing the nearest exit signs. The foundations for any IoV system should satisfy the following: authenticity of data, anonymity, integrity of data, confidentiality of data, low-overhead, and availability.

In Inter vehicular IoE, since we integrate many technologies and services in a heterogeneous environment, an attacker can malfunction many vehicle functions such as breaks and steering wheel that can make us realize the crucial requirement for better security solutions addressing many aspects of detection to vulnerabilities and remediation.

IoV is challenged to meet the traffic demands and energy supplies as many devices lack a standardization to do their work with homogenous sources creating troubling thoughts for the designer to realize this adaptation that

needs to be filled. For instance, RSUs demands battery storage and charging cycles and a robust control algorithm for integrating into the IoE scheme. The battery capacity affects its driving range; such fast charging might disturb the battery balance. In addition, the hand-off timing and the gateway that the nodes use inside the network, and the suitable technology to use [47], we can find different authors who talked about IoV challenges and identified different vulnerabilities, especially in V2V; hence, it's vulnerable to malware. More interested readers can refer to [48] that illustrate current IoV devices and services coming without integrated security features and how this is affecting the secure by design mindset. Ensuring security in IoV is a major concern and perhaps the most difficult in the development process due to that heterogeneous environment where a million connected things, services, and protocols are conjunct as most current IoV infrastructures are in their primitive phase. The absence of a security frame with enough computation power alongside ensuring the CIA triad so that only legitimate users can access the system power and nodes don't masquerade to keep the integrity of content from corrupting. Further, the long time cycles to produce smart vehicular models while giving security patches updates to the old ones drive the industrial manufacturers to see it as a waste of long-time because factors such as workforce and resources are at stake in this kind of process, in the same sense to ensure that nodes accomplish their assigned functions in the face of threat as data verification and impersonation.

6 Policy

The SolarWinds cyberattack in December 2020 announced by The Cybersecurity and Infrastructure Security Agency (CISA) and HomeLand security agency highlights the ease that systems can be targeted with the lack of better cybersecurity policies that governments and organizations didn't force yet. It will likely be a cyber-attack on critical infrastructure that will result in the loss of life as CISA described the attackers "they used advanced persistent threat (APT) demonstrating patience and multifaceted tradecraft in this intrusion" [49]. The question about how can we define our threats and set their priorities?

Demand a strategy that encompasses areas and actions as manufacturers and suppliers neglect proper security policies and many organization or users relies too much on third parties. As shown in Fig. 3. In policy, we need to address the fundamental policy issue, introduce a security-first mindset, and make governing bodies and organizations see that neglecting such policies represents a risk on the entities themselves and creates risks for the clients and partners. Enforcing policies and regressive compliance will

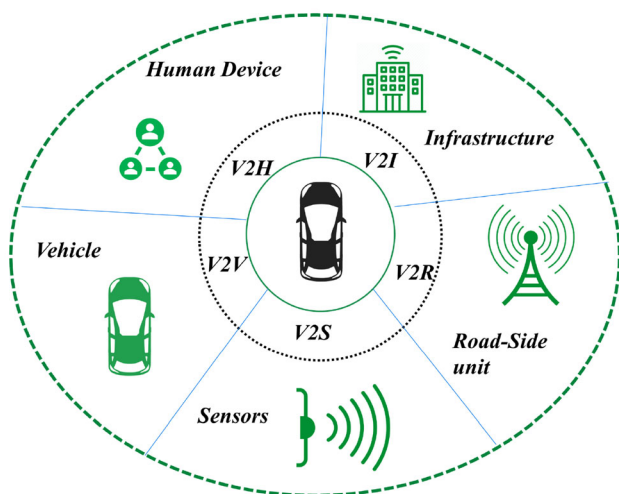
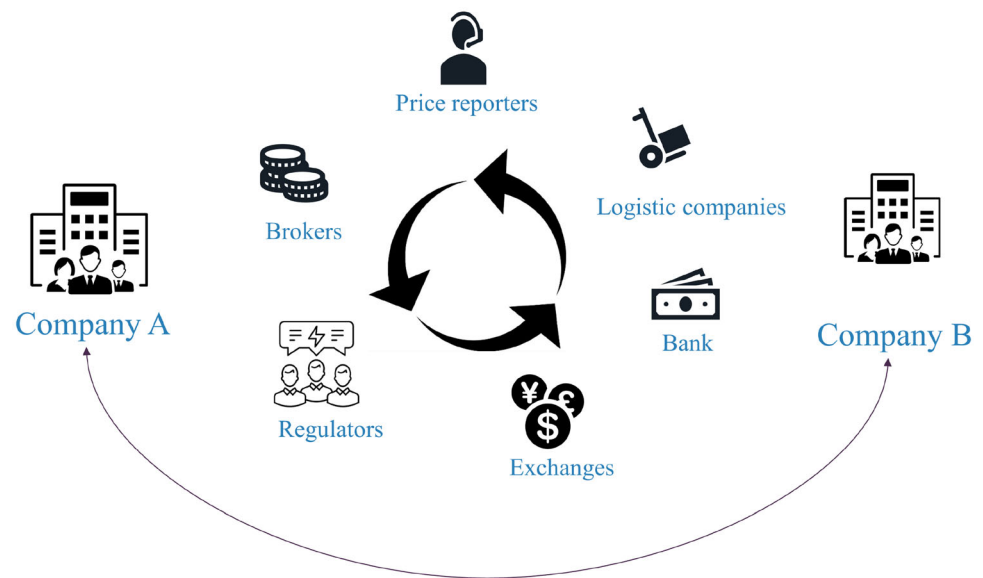


Fig. 2 Internet of vehicle communication

Table 4 Brief comparison between the IoV and IoE IoV

Internet of vehicles	IoE Internet of vehicles
Energy underutilization	Energy efficient usage
The charging stations are wired	Can be both wired and wireless
Low emission	Zero emission
Centralize charging	Distributed charging
Powered by batteries	Homogeneous power sources
Require energy management system (EMS)	Real-time machine-to-machine communication (M2M)
Battery charge through grid to vehicle (V2G)	Bidirectional energy exchange between grid and vehicle
Devices configuration are done manually	Plug and play
Waiting time for charging	Charge on the go

Fig. 3 Traditional energy transactions and third parties involved in

allow us to measure the days' baseline of attacks and insurance for the future while keeping evolving measures.

To investigate the threat and risk management system and conduct a deep analysis of the system with the given recovery plans for multiple scenario outcomes and signing quarterly on the threat posture of an organization will encourage the constantly evolving strategy to become more sophisticated with state adversaries at play. Such a mindset can make us identifying the essential services, which are too valuable to the system. Lastly, both the academic and industry sections should be keen on cooperation to understand that the projected risk and damages can include all of us to reach the finest practice standardization. Understanding that the evolving IoT interfaces will bring new challenges as these changes are driven by convenience and making things more software-based aims to grant the users improved control and cost reduction. However, it created security and latency problems because we have to ensure the quality of service and bugs. Using software-based interfaces will help update releasing and utilizing time-saving for developers, ultimately bringing profits.

Nevertheless, remote connectivity has changed the interactions between users and devices but presented a problem that didn't exist with physical interfaces called hacking, hackers are diverse and determined as the persons they attack; some are lone-wolves or work in teams. To mitigate hackers, we must think that our system is as strong and safe as its weakest link because they can rely on using social engineering tactics to manipulate the personnel into revealing confidential information allowing them to have access by any means necessary, ranging from following to blackmail or exploiting human nature. For those reasons, *cyber elasticity* is needed to perform more efficiently and powerfully; this resiliency as a part of the cyber-framework is mandatory to set the desired goals with the informed decision considering obligatory and awareness campaigns should be conducted regularly because, in the digitization era, we can't allow illiteracy near sensitive data. Also, there must be a procedure to provide traceability within the supply chain and imposed patch management life-cycle updates by teaching the humans asset that the training never stops.

Cyber framework, based on our energy system for articulating a fast response in emergency cases, this framework should realize our system vulnerabilities in advance as part of crisis management circumstances, and putting it into action should be fast. Its scheduled stages are to identify, protect, detect, and recover. It will ease our management, auditing and improve the protection; without it, we will have security gaps, hindering its attempts to gain its reputation and trust by public and affected associates. The demonstrated pattern in attacks prove that the more sophisticated the attacks, the higher value of the data or the organization. With this in mind, treating providers/partners with caution because it is trusted doesn't imply it's secure. The need to insert framework in designing is of paramount importance, and the best suggestion is to include:

ISO 27001/27002: ISO 27001: Also known as segregation of duties, is a documented framework to help make security management system drawn requirements standards as a bunch of good applicable practices; when implied, it can close the lagging gap between regulation and security as management falls behind. This stander addresses the implementation, revision, and improvement to view management distinctively to realize management goals and aims intended to achieve security accompanied by internal audits [20]. Where ISO 27002 is a set of practices used in information security as a universal code for security control with objectives relating to risk assessment considering the CIA triad to apply suitable control measures, for example, technology access control and physical access usage procedure and policies [50]. The defined stander addresses the information security policy, cryptography, communication security, operation security, compliance, human resource security, and maintenance with physical security in a comprehensive study.

Security Controls Center (CIS): This framework lists 20 critical controls that need to be deployed to achieve the best security environment. This frame draws the baseline map for systems configurations, divided into two levels: the indispensable basic security referred to as “cyber hygiene” to control hardware, software, privileges, and vulnerabilities, then the larger security environments referred to as “technical measures” that provides methods to deal with data recovery, malware defense, boundary defense, etc [51].

NIST Cybersecurity Framework (CSF): Standards and guidelines for both government and organizations that aim to implement the safest security practices. Based on five groups: proof of identity, defense, detection, response, and recovery process to obtain long-term security measures. It can assure security and what's it really adds is the consideration of cooperation with organizations that don't follow security regulations in an incredibly fragmented fashion. This fragmented frame includes online learning

modules, case studies, and guidelines, making it easier to implement for decision-makers [52].

NISTIR 8259 by NIST: It aims to eliminate the chaotic procedures by allowing manufacturers to improve IoT devices' security before sending it to markets and provides essential security functions that allow users to access related information that reduces the prevalence of compromise and attacks. However, not only the newly manufactured devices but the ones already deployed, the frame is separated into two sections: before the device reaches the market and after the device reaches the market. Safeguarding the CIA pilers and the customer interest [53].

The Endorsement of promotion, better remediation policies will fix the problem as early as possible by keeping an eye on the changes inside the energy market to gain an equipped set. Some tasks need specialized skills and training for the human factor. Likewise, including the security officers (CISOs) in the executive body will showcase a bigger range of influence and fast response instead of just reporting while supplying them with a budget for security purposes will make every decision consider the cybersecurity risks. Whether it's about product catalog, breaches, and deployment of technologies will be seen as a continuing protection phase displaying that cybersecurity is the core value of our actions.

Final remarks can be about the need for adherence, especially in coding, as coding standers supervise our work and can ensure the focus is centered on cybersecurity because compliance to standers will give the best practices and minimize the time needed for error detection to help in keeping hackers away and checking the program if its run in the anticipated flow and measuring the end of the device service time.

7 Threats in IoE and IoE related fields

7.1 IoE

Internet of energy in its core possesses poor management of its environment which is responsible for supervising transactions and energy efficiency, but the manufacturers and government regulations fail to prioritize secure by design mindset for such highly advanced technology and the see in the old traditional modules a fast solution since the end-user main concerns will be around the delivered service and agreeing to ambiguous license agreement they don't read it. This direction will only make the IoE inherent the existing deficiency because the majority of used security are uncoordinated combinations that can cause isolation in cooperation due to redundancy or possible conflicts

that a witty attacker can use as attackers try to find a weak link to exploit it. The inherited threats will create openings for vulnerabilities but considering new factors, for example, when controlling an electricity grid or a gas system; the aftermath and involved hazards after the attacks mandate the need for tighter security to prevent the hazards attacks damage [54]; eventually, we must realize that our security is stronger as its weakest link. Altering and changing sensitive data or manipulating the consumption states will allow fraud or overloading of the system. Also, the attackers can use malicious codes as a way of deceiving by infected codes to affect devices within the grid or the vehicles internet or to use SQL as the target is the database queries because many applications in the IoE uses SQL queries and their database relationship will permit some attackers even to be the administrator of the database [55]. To mention a few attacks, in February 2020, the biggest distributed denial of service (DDoS) ever recorded happened, attacking Amazon Web Services (AWS) and Amazon mitigated around 2.3 Tbps of data [56]. Stuxnet attacks targeted the Iranian nuclear facilities in the city of Natanz was the first time attack based on codes causing enormous physical damage by interfering in the centrifuge controllers that kept simultaneously sending false signals to obstruct the monitoring until the facility exploded, alternating the control logic and giving access to remote equipment. IBM conducted a study that found that the average total cost of a data breach in the United States has grown from 3.54 million in 2006 to 8.19 million in 2019, approximately a 130% increase over 14 years, and during the period from 2014 until 2019, the malicious criminal attacks had grown from 42 to reach 51% [57]. Figures 4, 5 shows the increasing number of attacks and the manifestation areas.

In this context, we can divide the threats into three groups:

The first group is in sensors and devices. Since most of them already have low specifications and many of them are built without supporting security patches or hardware compatibility features. Using ICT technologies helps to

support the bidirectional information flow by improving the control strategies to attain better observability and controllability in power systems [58], but the adversary can launch a false data injection attack (FDI) to tamper with critical data. Alternatively, making a crucial node as a sinkhole that can't deliver the routed data. In any spoofing attempt, the hacker will try to make an identical sequence number for the intended victim to give a false sense of trustworthiness to impersonate an entity.

The authors [59] showed how an adversary could use IoT devices to launch attacks on the grid which has been classified as a new type known as Manipulation of demand via IoT, validating the result with simulations and how they can cause large scale blackout depends on the scale of the attack along with modifying the cost to benefit third parties in the electricity market. The manuscript also shows that high-voltage IoT instruments such as air conditioners (ACs) and heaters could be weaponized for launching a large-scale attack on the grid.

In [60] the author demonstrated that the personal electric vehicles attacks and electric vehicle charging stations could cause disturbance in the grid, jeopardizing the grid operation by the use of the openly available data to tailor a data-driven attack using partial eigenvalue relocation. If supposed, the collected sensor data has been altered. In that case, it will falsify the system reading and the system-dependent parameters. In some cases with a highly critical system such as NSYEME (Integrated System for Emergency) that operates the pervasive grids to predict disaster evolution stages, manipulating sensor data will guarantee a catastrophe can make the facility explode, leading to enormous financial damage and human life loss.

The Second group is in the network side carrying the connectivity and data exchange among utilities for communication and it can generate opening for attackers to disrupt energy supplies such as congestion attacks and collision attacks to tamper the information. In the network, all the data gathered by the sensors should be utilized for the system benefits to prevent resource wastage as this huge volume of data is transmitted to a control center using

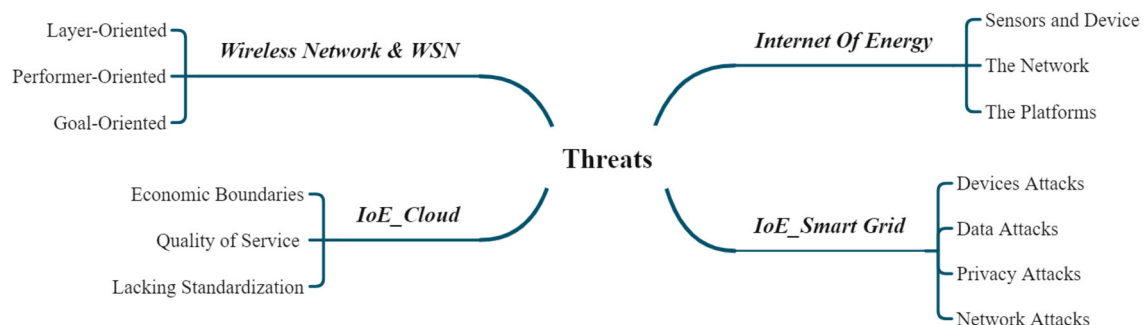
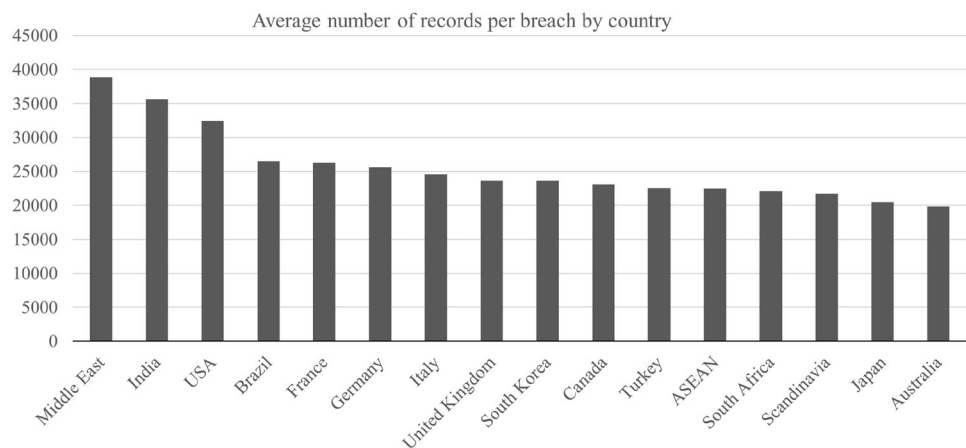


Fig. 4 Threats in IoE and IoE related fields

Fig. 5 The increasing attacks per countries

IPv6 wirelessly as a transmission medium by the bidirectional exchange that will permit to control these physical objects during the course of the operation. However, the attacks on wireless and connection structures fall into three categories: goal-oriented, performer-oriented, and layer-oriented assaults [58]. Goal-oriented attacks can be passive that monitor the network without interfering and observe the data, such as eavesdropping that normally occurs with poor endpoints security to steal the information from the connected devices during transmission, which can be a prerequisite for a coming attack after gathering sufficient data about the system. Active attacks will differ because they interfere by crippling the network activities and opened sessions through worms or denial of service. A worm attack named silex infected 2000 IoT in one hour; this worm attacks delete the firewalls saved rules and remove the devices configurations, and destroys the IoT devices storage capabilities randomly to any mounted storage that cross it, finally, delete every device settings and flushes the IP tables before halting the operations [61].

Individuals make Performer-oriented attacks towards another individual to steal transmission channels and responsibilities by injecting false data (FDI) or man in the middle. The hijackers spoof the identities of the nodes that are involved in information exchange that can make the transaction becomes valid events during transactions. Layer-oriented attacks target a specific layer instead of the whole network by contravention communications protocols, harming the physical layer, or attacking the TCP/IP stack. These attacks want to strip the layer from its function or be in denial of sleep to cause energy depletion and flood it with false data [15].

The third group is in platforms for services. Most of them originate from open-source codes, so they are accessed by many people who can see the code structure. The IoT platforms purpose is to reduce complexity in operating a system and provide actionable intelligence for the operators to simplify deployment, but the complexity of

such an environment unintentionally produce vulnerabilities that can be used for security breaches in connectivity, programming constructions, security, and management to help the single stand-alone device to be integrated into a system that facilitates interoperability that should aim to secure end-to-end service delivery. Connectivity is the cornerstone for such platforms, and no platform exists without it that bridges the gap between data acquisitions, remote access, and control points. A trustworthy and secure platform is the ultimate goal to achieve that users rely on that manages the devices and provides application logic to handle the devices and the data in a management frame. The security role in the platform is ensuring identities, providing privacy, and secure the CIA triad across all components. For example, a DOS attack is meant for a broker by flooding the communication with false messages to consume resources and degrade the performance as clients keep making requests. The server can't respond to the request also suppose the user wanted to log into his account; it demands that the server checks the database for the credentials and send it back, but because its resources are spent by false inquiries disproportionating the supply and demand balance it gets overwhelmed and denies legitimate traffic.

In [62] the authors suggested a platform that supports customizing IoT platform programmatically for logic via trigger-action programming (TAP), so the platform itself possess automation and facilitate the understanding even for nonprogrammers named it IRuler that utilize surface mount technology (SMT) and takes the metadata and the deployed configuration as input and replies with representation for IoT deployment, after that the deployment gets checked by an engine to detect vulnerabilities and the user can receive messages about the severity of the situation according to misconfiguration mistakes to produce a trigger-action platform.

7.2 IoE-smart grids

As every grid component carries vital information, many devastating attacks can target the main grid, smart meter, distribution pole, wireless communication aiming for the physical entities [63]. Supposedly the attackers possess pertinent information on the system layout. In this case, they can affect the energy production and manipulate the operational data, such as decreasing or increasing costs by relying upon false data injection (FDI) [64]. Invaders or insiders will try to access and influence grid devices and components such as sensors because IoE grids have remote-based configurations [65]. We can have four types of attacks and they are almost driven from IoE groups:

The first type is device attacks, which aim to compromise the device as devices become attractive for the mass market. Usually, it starts before the main attack in the form of malware spreading disguised as a legitimate program, for example, a trojan horse. The developed software will spread across the devices, infecting the grid by a malicious code or mistakenly downloading it by a coerced user aiming to alter device functions or sending information to a third party. In addition, the grid is a compound combination of information and communications technology, processes, and individuals who can allow the attacker to access the power management unit (PMU) contents or the sent packets by the smart meters to sniff its information with the lack of encryption mechanism a bigger chance for observing serious information occur [66]. Well-known attacks include side-channel attacks that try to obtain the cryptography keys using power analysis or electromagnetic waves; power analysis investigates the variation in a device consumption rate, formulating an idea about its contents by inserting a small resistor to measure the voltage drop. Electromagnetic waves attack uses the electromagnetic spectra measuring the emitted radiation followed by signal analysis using a sensitive probe [67].

The second type is data attacks, as the assailant aims to mess around with data by altering, deleting, or inserting false data, which can mislead the grid; the grid control system will keep a lodger about the activities in the grid side, voltage and sensors values, and commands or the network side with the logging and credentials. Some customers will also try to change their data states like utility bills and manipulate the database to escape paying high utility bills; this can occur if the database management was poor and not configured appropriately to exploit the system to his benefit. The false injected data (FID) will manipulate the original sensors and energy consumption data, causing cyber terrorism attacks that damage the revenue stream or hold a financial benefit to another party. FID mostly happens in state estimator and smart meters for altering the

estimation of the power systems concerning the phase and voltage measurements by that the monitoring functionality will be destabilized leading to untrue operations impacting the grid stability and daily routine process, one of the predicted consequences for such action is a total system crash damaging the grid infrastructure as well compromising the entire integrity and availability. In addition, smart meter manipulation will cause the sub-systems and wrong phasor information used with replay attacks so the generated energy can be routed to incorrect locations causing serious physical damage or the potential of false control commands.

The third sort of attack will be intended in the direction of the private sector. In contrast, the attacker tries to steal personal information or unauthorized analysis of information or users credentials for the system or credit card information that should be kept secret merely between concerned parties. This type of attack normally will be planned together with the insufficiency in caution to handle a huge amount of data compromising consumers' privacy, safety, and security as privacy must be kept integral to sustain clients' faith, and the gained assurance will help the grid acceptance. Some used attacks mechanism include password sniffing and guessing attacks, social engineering, and attempting to read a device memory that greatly compromises confidentiality and integrity and can reach appoint of denying availability if it changed a device control function. Crafty attacks such as spoofing by the deception of devices will target the PMU and sub-stations; by doing so the fake broadcast packets will spread across the grid, shutting down the gateways and devices and harming the availability of the grid.

The fourth type the perpetrator's goal will be the network by any means necessary that can suit his agenda by trying to denying the availability features or overwhelming the grid with a flood of fake information [54]. Likewise, tiring its resources or phishing where he pretends to be a trusted source and requests information as a legal source and this type of attack can cause detachment between the grid entities, especially in disruptive systems where some parts can't respond or collapse. Network attacks will compromise communication kits such as multiplexers. Besides, the network has layers- the physical layer assaults can be as jamming devices that stop network availability by connecting a jammer in the system or jamming wireless communications [68]. The usage of TCP/IP stack in the grids can allow a vulnerability to be used by denial of service attacks (DOS) by that the denial of transmission or blocking it as the attacks will nullify the resources utilization. DOS can take another form by draining the system components such as memory and bandwidth which adds another burden to the grid that already has limited computation power. In the transport layer, availability is

commonly compromised by the interruption of the end communication so the traffic will not be received; the intentionally architected wiretapping will be a massive security breach as networks have routing tables and this traffic travels similar paths [16].

7.3 Cloud-IoE

This section will introduce the benefits of the cloud to IoE and demonstrate the security risk surrounding it. The principle of cloud-based IoE offers the technical conditions for integrating electric and information infrastructure that connects the users and coordinates the engagement to adjust energy capacity per energy consumption and generation for technologies like IoV and smart grid, which are now matured enough for the cloud. A huge advantage of the energy cloud is allowing the substations and central stations to communicate directly with the cloud instead of the previous existing difficulty in energy management communication exchange and allowing the grid to store the extra generated energy and route it to higher demand location and minimize the load while engaging the customer in the energy-saving process [69]. Smart grids can be seen as cyber-physical systems that tie the electricity systems and cyber-infrastructure with the presence of internet connectivity as a medium. The increase of entity integration in this physical system can increase security concerns in three aspects: data confidentiality from leakage, data management and monitoring, and user privacy in terms of authentication and authorization. The cloud role in this vision will be providing servers for decision making to integrate the grid client, enhancing the security in the flawed communication method, data storage security, switching from all in-premises defense, and assisting by cloud-based privacy scheme. This fragility caused the developers to use cloud-based IoT platforms because the platform enables upfront management and computerization of connected devices within the Internet of Things universe. In a manner of taking the problem of breaches to the platform developer side.

Hostile activities such as DOS in the cloud for the smart grid can be done by generating a huge workload on the target to keep them occupied and preventing them from reaching a lower power usage, forcing them to operate with a 22% reduction of power utilization forcing the system to operate at the maximum frequency speed, voltage and overheating. In today's architecture, the modern servers relate to all these mentioned operating features, placing a burden on the CPU, memory, and disk in the data centers to absorb more power and increase the budget [70]. Forcing the data centers to operate at their maximum constantly or continuously fluctuating the spinning of the hard disk spindle engines will drain additional power. At last,

distributed troublesome attacks can trigger the anti-virus to waste more CPU capabilities and drain energy.

Autonomous vehicles mainly process data from cameras, light detection and ranging (LIDAR), radar data, ultrasonic sensors data that work relentlessly to capture data from the physical world in a data pool for vehicle operations. To meet these storage demands, we need highly efficient storage centers to perform predictive maintenance, monitor traffic flow, and possess consoles that contain GPS navigation [71]. The inter-vehicular cloud can be achieved by one of three schemes: Temporary cloud based on demand for registered members as vehicles temporarily seen as part of the infrastructure to accomplish a certain task or sharing resources. A permanent cloud to be accessed by a user for communication between the vehicle RSUs and the cloud, only the authorized vehicle that established the connection has the right to do so and share information between the connected vehicles and the data center. Hybrid cloud architecture by combining temporary cloud and permanent cloud architecture [72]. The inter-vehicular cloud can face threats like certificate replication using fake certificates and other users' credentials for authentication purposes and eavesdropping to access confidential data of a vehicle such as location and user ID. Jamming attacks to interrupt communication and spoofing attacks can generate phantom and fake traffic in the sense that there is congestion.

To be brief for the cloud itself, the economic boundaries, social worries from users' trusting side, and administration concerns are the key concerns. However, precaution must be taken when dealing with threats such as hijacking and session riding. IoE heterogeneity versus cloud capability also throws its shades on the security challenges since the absence of a standardization architecture for the combination of both IoE and cloud as the usage of different operating systems, devices, and applications [73]. Moreover, adding a new node to join the network and the identification mechanism to ensure the quality of service (QoS) across the structure because without QoS, our structure will crumble and fails in situations where the data can't be quantified or if the output not accurate which can generate latency due to IoE time-sensitive nature.

7.4 Internet of vehicle

The desire to have autonomous cars that self-park or route was driving the IoV but the different routing protocol in IoV was focused on homogeneous models instead of heterogeneous models [47]. Thus, producing vulnerabilities that act as an opening for many attacks such as false data injection, allowing assailants to forge the battery status or temperature influencing the driving decisions or tricking

the global positioning system (GPS) targeting the integrity of the data. Cybercriminal minds always playing on vulnerabilities cord with that to say using the FDI attacks or eavesdropping, where the routers/nodes delete the messages instead of forwarding them, creating a confusion statues resulting in a denial of service (DOS) attacks/spoofing attacks [74]. In IoV, GPS data has a significant role in several applications like navigation tools such sensitive information can cause physical damage if misused. The attacks that target the availability criteria are masquerading attacks and denial of service; masquerading will create a chaotic status and nodes miscommunication by pretending to be a trusted source, making the vehicle misbehaving. It can also block the routing process. Further, DOS poses a threat because it will prevent the authentic user from accessing the services provided, for instance, sending a jamming signal amongst the nodes or between the node and the roadside unit (RSU) can make the node drop information related to guidance or internal status to put the system in an unresponsive mode. The cornerstone for IoV is the network connectivity and any miscoordination or excessive latency can severely degrade the performance. Onishi [75] described the cybersecurity challenges in vehicular communication schemes by limited connection abilities that lack updating, poor computation performance that focuses on the car endurance abilities for working conditions, infrastructure that posses entry-points for database or remote connections that don't consider new attacks by built-in security features. Mitigation techniques for vehicular communication will later be discussed in section 8 & 9 as part of future mechanisms for the prevention.

7.5 Cyber attacks and assaults

Figure 6 shows the most common attacks faced by energy internet.

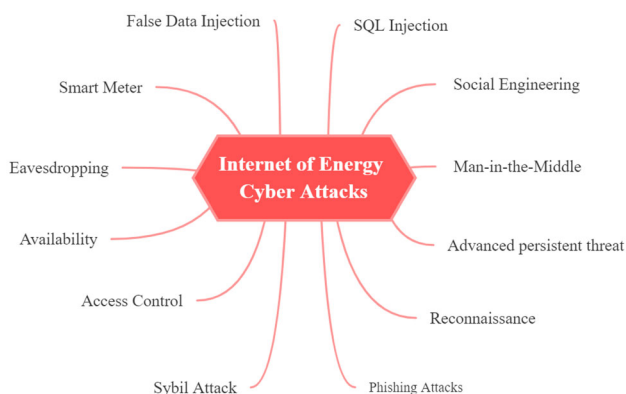


Fig. 6 Cyber attacks against Internet of Energy

7.5.1 SQL injection

Its main interest is the data-driven application by sending malicious SQL inquiry (script commands) towards the database as plenty of IoT devices are accessed remotely so a firewall can't protect them. When entering these statements and it gets executed, the attacker will try to get sensitive data like credentials and permission. Upon a successful attack, the attacker will reach the database contents and extract a huge data portion [76]. The usage of SQL injection assists the hacker in establishing a foothold to start controlling the system. From this point, they can access the devices and modify other users' privilege levels eventually, controlling the entire network. In the case of a smart grid, it may result in a disturbance that will lead to a blackout as the hacker will access the smart meter stored data in the database, multiplying its severity on the network as the deception will continue making it difficult to determine the trusted sources from the malicious ones. In wireless networks, the Sybil node will generate different identities, and the Sybil node can communicate directly or by using a medium node with other nodes by spreading the scripted commands damaging the whole system. Usually, a Sybil node will declare itself a cluster of nodes to affect the topology and consume an immense bandwidth. Due to this, the internet of vehicles will also be attacked as vehicles always access the connected infrastructure. This deception of vehicles can cause private data leakage by the forged identity or broadcasting false messages about traffic jams and declaring false positions. Author [55] has provided a method to avoid Sybil attacks on wireless sensor networks (WSN) uses a distributed random password comparison (RPC) algorithm that makes a routing table (R-Table) that holds each node's id, time, and password then the route through the intermediate nodes is identified and compared with the database to detect Sybil nodes.

7.5.2 Social engineering

The attacker interacts and manipulates IoE system users psychologically to accomplish his objectives to get delicate data by throwing bait and hoping the user will take it and act upon it or send an email from what seems to be a trusted source contains inside it a malicious link. Forms of severe attacks can be as blackmailing for employees to give sensitive data about the organization [77]. In smart-grid, sending fake queries and the user didn't verify them can cause malicious software installation. One of the famous social engineering attacks is phishing and can be described as one of the first steps to put an entire entity at risk. Hackers can contact the customers or the staff with little obtained information from receipts and try to obtain more sensitive data or send a spam email link to a malicious

website that will demand them to enter the logging credentials knowing that many individuals use the same credentials for many sites because it will be easier to remember. Phishing is a type of social engineering or a deceitful attempt to obtain sensitive information by disguising and decoying the intended recipients. Varying levels of complexity, phishing attacks can be on a mass distribution scale or toward individuals [78].

7.5.3 Man-in-the-middle or person-in-the-middle (PITM)

This attack aims to hijack communication by eavesdropping or intersecting and intercepting all the data between two parties as a third, invisible person [79]. In a scenario, when sending data between a sensor and a microcontroller, it can be altered as a consequence, we lose data integrity and data integrity attacks like MITM exploit the weaknesses by hijacking or corrupting the processes in the smart grid, the attacker can harm the data communication between the smart meter and the station units by using MITM to modify the data packets. MITM ultimate goal is to manipulate the messages, especially between centers and the grid or the vehicles, to affect system accountability [16]. This attack will grant its user the ability to sniff LAN by address Resolution Protocol (ARP) poisoning to impersonate another host IP address and associate that address with his MAC address to disseminate false information to entities and get privileges to change the parameters illegally.

7.5.4 Advanced persistent threat (APT)

It is called advance because it uses advanced equipment needed to keep the attack going and the phrase persistent. After all, the hackers are determined to achieve their goal and for doing so they need to be evasive to avoid detection by the detection system. Thus, the higher threats provide a more valuable reward which is the sensitive data or components. Persistent threats evade detection mechanisms and propagate themselves around the systems to infect more devices making the danger more physical than those other arbitrary programs. APT is a well-funded type of attack that can be a state-sponsored attack or a non-state-sponsored attack, but what is common that the large scale of it and its complexity and the determination to reach the precise goal, not like other cyberattacks that hope to find weaknesses accidentally. However, it takes phases to achieve this even if it meant that the attackers could remain passive for a long period or to mine the information [80]:

Initial Compromise: Infiltrate a target organization or system using vulnerabilities in the server or the system that the penetrator stumbles upon them by espionage tactics or by entry point.

Reconnaissance: It aims to collect vital information like subnet, host-name, active ports, and vulnerabilities [81]. It has two types internal or external reconnaissance. The internal reconnaissance is done on-site by carrying it out in the network, and mostly this method involves software tools as the attacker interacts with systems collecting information about the system and views internal network files. External reconnaissance is done without interacting with the system by making a human collaboration, including hackers' aid in finding information. Operation sharpshooter in 2018 was a detected framework to hack global infrastructure that relates to the energy section that included nuclear facilities and energy companies in the form of espionage; it started when users received files by dropbox and after opening them, they install an embedded code in the system that resides in Microsoft word memory and start gathering data for the benefit of the attacker [82].

Establishing a foothold: Denote the successful entry efforts to gain more access, reaching a state of more controllability over devices, and expand his presence by creating a backdoor to make an outbound connection to gain access to confidential data, weaken critical components, and extract this data without being detected using white noise tactics.

Escalating privileges: Obtaining the credentials and resources entree and the final goal of this phase is to manipulate the administrative privileges. In this phase, methods like password harvesting will be used. After that, a new backdoor will be established to remove the system accessibility so the attacker can take what he wants and cover his tracks.

APTs are regularly undermining IoT systems due to their complex interdependency and hardware security insufficiency is bringing multiple malicious activities, and assigning semantic detection tasks to each IoE device seems impracticable. Targeting the energy sector, let us say the oil or gas industry involves different service providers and third parties from independent to state-owned, which can provide multiple entry points for the whole system that now deploys new technology like drones and cloud connection that create additional vulnerabilities for a geologically dispersed set-up. As a group of skilled hackers with collaborative efforts tries to penetrate the system, they need an infected USB to launch an attack, whether it's by social engineering or zero-day malware [83]. Recent attacks that used zero-day malware or GreyEnergy affect programs adversely because the system can hardly detect them, but their nature they utilize awareness. After creating a profile about an organization that includes the software system and identify mitigation capabilities were reversed to their advantages. GreyEnergy, for example, targeted the facilities that use SCADA software and compromised the web servers and internal networks and then claims to be a

trusted source that sent malicious emails to redirect all the requests to a server owned by the attackers [84].

The author proposed a novel cyber-situation method for IoT schemes built upon zero-day attacks activity to recognize the attacks involving an edge computing framework for typical IoT systems accompanied with a module for describing the attack actions. Furthermore, the proposal of a command and control server for domain name server (C & C DNS) to identify the harmful activities where the normal IDS have a deficiency to detect by digging these activities that don't consume too much computation power. Further, a mature access-control mechanism to enhance the defense capabilities [85].

7.5.5 CIA triad

In IoE, when obtaining access to user information without his permission and sensitive data exposure. The assailants will attempt to access communication information by spying or snooping in illegal ways as eavesdropping to damage the secrecy of data and intercept legitimate traffic flow by this way. Eavesdropping also damages the integrity and availability because monitoring the system can reveal the structure or help to predict the behavior and this attack is very dangerous to IoE because it's passive and doesn't disturb the original data. The integrity can be compromised by false data injection, such as sending packets to falsify the original data like meter data, utilization, vehicle routes, and emergency response [86].

In the smart grid, availability attacks will undermine the grid operation because of the difficulties in accessing the grid and its components, mostly in SCADA, communication channels, and distribution management systems (DMS). It takes the shape of DOS that interrupts, floods, and delays the transmission abilities and aims to overload the grid so the system functions cannot perform appropriately and losses the legitimate packets. The availability can be compromised using jamming to block the communication lines by that the smart meter losses its functionality and the grid power distribution, making it the most dangerous attack for a smart grid. Integrity can be done by tampering with the data during the transmission or corrupting the original data resulting in a loss of integrity, leading to incorrect decisions and disrupting the critical data exchange. Confidentiality is a privacy violation by manipulating the access and violating the end-user and the energy provider agreement that reveal private data as this private data is the customer data such as living addresses and company secrets.

In IoV, dynamic changing the network topology increases the difficulty of sensing attacks, and forging identities to gain disproportionate control over the network affects the network's data integrity, resources, and

performance. Most of the user messages on inter-vehicular internet are exchanged publicly, largely in the vehicle-to-infrastructure exchange (V2I), so the secure-by-design scheme has the utmost necessity for the confidentiality criteria. The integrity must satisfy the protected exchange of messages to introduce reliability and precision to mitigate deliberate or unintended modifications. The loss of integrity will compromise the service and produce a vulnerable system that can be affected deeply by jamming, FDI, and message distribution. The availability criteria role allows accessibility and usability for the authorized users as units such as RSU are vital for collision warning. On the other hand, the vehicular cloud can be predefined to provide access control based on levels, but overlooking this will makes malicious vehicles access other vehicles' information. The security of the communication will deliver safer and effective vehicles with a method such as certificates revocation to protect the data as soon as detection is noticed, the certificate will be revoked [87]. For mitigation and protection, the CIA in IoV will be discussed with practical solutions in the following section.

In the cloud, the CIA permeates all the activity in the energy cloud to provide security and privacy to users, which is more challenged by data loss, data privacy, and unintentional credentials leak. Confidentiality in compliance with regulation to secure the data and ensure the concealment of information from unauthorized users means that resource-freeing attacks lead to memory leaks. Availability in ensuring the shared resources pools can provide the needed resources as the attacker will attempt to incite changes in resource exploitation because cloud efficiency is coming from multiplexing several client workloads into a single physical machine which will cause the discarding of messages, wrong scheduling, and ignoring requests that impact the performance. Integrity attacks such as migrant attack are made by generating useless resources reports about consumption or false report about some requests have been correctly performed mislead the energy cloud operation and monitoring role which will provoke migration processes compromising the integrity and wasting the cloud resources and worsen the expected performances [88].

8 Mitigation of attacks

A plethora of countermeasures can be used for attacks, but we will enlist the most efficient ones that hold the utmost importance for a security sector by referring to the mitigation tactics. They will be divided into two sections: before the attacker accesses (proactive cybersecurity) and after gaining access (reactive cybersecurity).

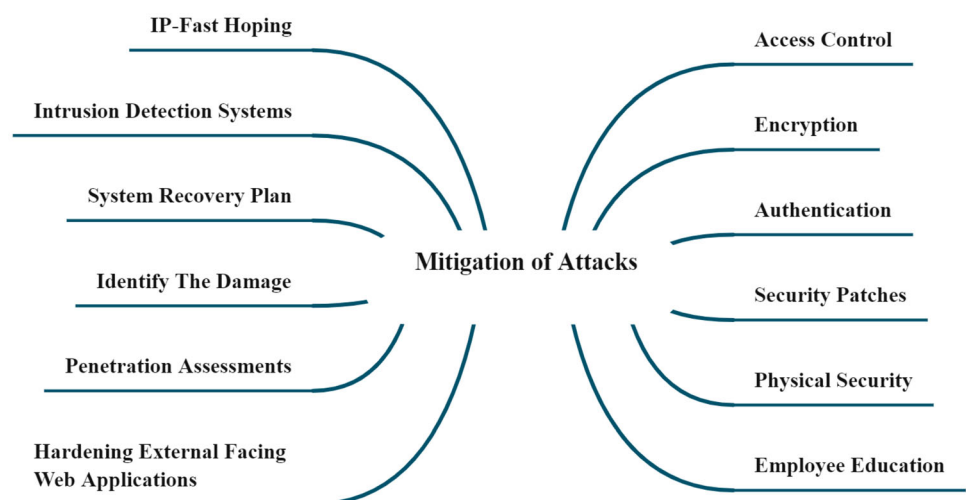
Proactive cybersecurity identifies the likely routes that an attacker can take to perform attacks, empowering our understanding of security issues. The first step is identifying IoE assets and identifying possible threats based on the attack vector cost, complication, reputational influence, and repeatability. Ethical hacking can add countless advantages in this case as it involves enumeration and scanning. Enumeration is recording all the identified resources in a system and discovering open doors in a system as part of the penetration test by gathering information through active connecting. Scanning aims for vulnerabilities discovering in ports, perimeter defense, and various system parts and both will be used in any attack initiation [89]. Figure 7 summarized the used mitigation scheme.

The testing framework should be present as part of the security strategy to assess potential devices against the security plan and be used after passing the test and third parties will be listed in a greenlight list or banned list after investigation. Employee education is crucial because human errors can lead to a catastrophe or helping to produce more vulnerabilities and ignorance was always the hacker's best friend. The usage of multi-factor authentication (MFA) to elevate privilege and deny forgery from gaining access. MFA demands represent more than one piece of evidence for the identity, such as biometrics and additional questions that limit the data accessibility to be only by authorized users and prevent the risk of loss and identity theft. Physical security is an important aspect to prevent tampering with data which can damage our system gravely. Therefore, a tampering-proof mechanism should be employed and integrated into both the grid or the system by applying pentest. Auditing and testing by penetration assessments to check and validate the security to ensure safety even in changes that occurred on the system by exploiting the vulnerabilities and simulate our system

response defining its effectiveness level. Performing pen-test routinely will direct to weakest links in the hierarchy and measure the resilience of such a body and gathering vital in-depth information will help to prioritize the probable weaknesses. Physical security can be employed by defending against the undesired logical access, facility design, surveillance, proximity readers for access, controlling the employee, and incident response plan. The outdated on-premises security only can't cut it nowadays, so the need for off-premises such as cloud security is growing by days accompanied by an incident plan to minimize the possible impact that covers the team response and what can be done for isolation. Patching the infrastructure regularly and installing updated software versions will add the capability to upgrade the future firmware and neutralize the vulnerabilities. A practice to increase security since the web application can impact the software and hardware configuration is hardening the external-facing web applications by imposing the role-based policy (RBAA), outlining encryption policies, enabling firewalls, disabling unnecessary ports and services, and using a network access list. Similarly, both anti-virus and anti-spyware should be active and updated regularly, along with the site-to-site VPN tunnels the entire connected network in one location, adding an extra security layer to share the resources between the branches.

Simulating the threat by a machine named sandbox environment as a testing location to execute codes that are susceptible of carrying malware and monitoring its physical behavior and hardware providing visibility to the structure and emulating the operating system to discover malware in early-stage, and the simulation-based report can be used to harden the system [90]. Another security practice would be implementing time-division-multiplexing (TDM), which dedicates time slots for nodes to transfer the

Fig. 7 Attacks countermeasures



packets to permit the nodes to have time portions to access the channel.

In [91], the authors proposed a rule-based anomaly detection system (RADS) to monitor and spot Sybil attacks aiming for sensor networks. The proposal depends on an ultra-wideband detection algorithm to work in a scattered way in which every node can trigger alarms that detect attacks regarding the presence of a central management body to defend against an adversary by isolating the Sybil node and each compromised node.

The use of routing protocol for low-power and lossy networks known as RPL to secure message transmission and its consistency check feature will guarantee the security in routing. Weekly and Pister [92] introduced mechanisms used for mitigating sinkholes/blackholes attacks that work by verifying the topological ranking using hash chaining named parent fail-over and rank authentication that demonstrated high results in practical application and increased the router density within the network. The authors proposed a framework to close the standardization gap that IoT security face by outlining the standard security protocols with Constrained Application Protocol (CoAP) for the constrained devices that achieved a stander management interface that eases data communication. The framework uses Datagram Transport Layer Security (DTSL) for communication protection. This proposed framework can secure the devices' commissioning and configuration and reduce the complexity of installing embedded systems [93]. The authors developed a framework for cybersecurity for energy internet with an identifiability mechanism along with a secure communication protocol and energy management and it has been evaluated based on game theory. What stands here is that it's a light weighted scheme that helps exchanges keys in unicast, multicast and broadcast communications. In addition, the framework certifies the security and privacy demands based on events and has been proven to provide robust results against a variety of attacks alongside a correctness behavior by Nash Equilibrium [12].

Creating a backup plan and training on this plan to guarantee data restoration as a recovery strategy or under attack scenarios, so panic doesn't control the employees. This restoration point should contain our precious data and configuration, and we can store it offline and place it in a tough place to access, and we can change or update it depending on the assessment. In addition to monitoring SQL statements to identify rouge statements, it attacks the application layer by applying measures that include tightening the access to remote users, checking the sequence of data/ tokens by positive pattern matching, and examining the code for security weaknesses known as static code checking. As mentioned before, authentication is an identification technique like giving any device an ID and

validating what it can get. Since this is the first step in initializing communications between devices and IoE, it's essential to the security structure in order to implement authentication; an implied deny policy is valued to control the access.

8.1 Access control

Fragile regulations have always been a vital nerve for hackers to use because it eases their job to find and utilize weaknesses. However, an access control mechanism for resources by hierarchical order or by using approaches that will be a barrier in their way will make the intended body for security acquire many benefits by setting permissions. Acquiring a robust authentication can ensure confidentiality as information will only be accessed by approved personal, which can relate to preserving the integrity as the information can't be altered.

Discretionary Access Control (DAC) means to restrict access by changing attributes of objects if you are capable, then you can access, such as setting a password in files or groups in here the users will grant these permissions by being part of the Access Control Lists (ACL). However, its limitation is failing to distinguish between human users and programs or computer bots as the users keep doing repetitive actions.

Mandatory Access control (MAC) a hierarchy built on security levels as predefined regulations by the administrator. Users can only access resources that are equal to or less than their security level. It has high sensitivity and requires a clearance, and its cons that it requires periodic attention from administration and users have to ask approval for their data parameters because permissions will be granted only by the administrator.

Role-based Access control (RBAA) taking MAC and DAC concepts to implement both, is the most advanced access control method to access what is only necessary and enough for you to do your job. Because the users get assigned to roles and the followed policies will grant the rights of these roles.

Andaloussi et al. [94]. Proposed a model that grants the least privilege and more excellent adaptation for access requirements claimed as a feasible method for deploying even in the existence of resource-constrained devices uses the concept of capability (DCapBAC) as keys for accessing and permissions that fall under a set of privileges, so for accessing an entity the user is required to send the key along with the request. Therefore, the receiving entity will know the request and the key with access to evaluate the request.

8.2 Encryption

Encryption is taking pieces of information and scrambling them so that they can only be readable in a certain way. For encrypting our data and making it encoded comes the need for a unique way to decrypt it to be readable again. If any unauthorized person tried to intercept the data stream, he would only get a gibberish code that doesn't make any sense. The encryption method for scrambling the information is known as a cipher, and there will be a logical key to decipher the information again, and the bigger the size of the used cipher, the more the combination possibilities for the encryption. The most common are: symmetric, asymmetric, and hashing algorithms. Symmetric encryption is using the same key on both ends and asymmetric is using different keys.

On the other hand, hashing makes the data (plain text) pass through mathematical operations done by an algorithm to produce a special message that verifies data integrity. Algorithms such as Blowfish use symmetric encryption that uses 64-bit length, AES (Advanced Encryption Standard) is a symmetric algorithm that can use up to a 256-bit length, and whirlpool hashing algorithm that uses 512-bit are very trending in usage for encryption purposes. Homomorphic encryption can encrypt the data during and after computation, making it a type of encryption with computation capability when a third party or a server requests information by sending it in an encrypted state. The received result will also be encrypted, and the endless number of operations that can be done on the data while encrypted is considered the most significant advantage of dealing with the ciphertext arbitrarily, even though any function can be done in this context [95]. The fully homomorphic encryption has a promising aspect by expanding the homomorphic encryption by allowing many different analytical operations on the encrypted messages. It will facilitate information sharing across traditional boundaries without sacrificing any method used, which is beneficial for cloud-based solutions. Furthermore, in the cloud-while making parallel computing with FHE, it will be an enabler for more operations on the cloud side so that the tasks can be divided at the same time by nodes which holds more security, and the FHE has been proven to decrease the processing time [96].

The author [97] proposed a solution for node privacy in IoV infrastructure by using homomorphic encryption with a fictitious name to carry communications with a chosen frequency. However, the alias name changes during vehicle mobility because while the vehicle is connected to the infrastructure, the vehicle information will be public, such as locations and speed rate, allowing both the physical and logical addresses to vary by time. Hence, specific vehicle

data and message transmission will not be recognized by the network in a way that two alias can't match at the same time as FHE scheme was implemented, and this method was proven to be efficient to reduce the communication overhead and better-quality security. Additionally, a re-encryption scheme by FHE is deployed to improve the encryption abilities that consider the vehicle mobility and minimize the computation time so the FHE provides dual authorizations.

8.3 IP-fast hopping

A software solution for denial of service (DOS) attacks in the network layer its principle is to hide the packets and their destination and the session information by a pool of IP addresses that have the location of many routers, making it very hard to detect a packet IP. The encapsulated real IP inside the imaginary capsule is scheduled to change in real-time [8]. Dos can be considered the main threat for IoE that can affect its components profoundly from functioning. Therefore, implementing IP fast hopping will be very efficient for the system overall security by giving clients the power to conceal the content of the communication and the destination of the packet as the server IP addresses will change instantaneously in real-time by using the pool IPs.

Final remarks, embedded software module presented for enabling data protection to have flawless security called security-as-a-service (SECaaS), it provides management for both hardware and software with an optimized response for detected threats with the aid of highly experienced experts to replace the need for internal teams and services, which can analyze the security statutes, preventing data loss, prevention intrusion, monitoring, and response encryption. Promoting similar endeavors will open the door to secure by design development to address the security in the design stage, ensuring lower costs and saving the troubles of adding security after deployment or after breaches. The unaware victim will not recognize a compromised IoE device as hardware-based protection alone isn't sufficient. Solutions such as applying cryptography are better, but it adds computational complications. CPU and Nano-scale Memory development will balance the security, cost, and performance requirements while switching from the old IPV4 structure because of limitations to IPV6, which supports numerous objects and deals with mobility and scalability necessities.

8.4 Methods if already access to the system occurred (reactive cybersecurity)

This section tries to contain the problem by removing all files installed by the attacker, separating affected devices, and fixing the wreckage to apply the recovery plan that

prevents future breaches by tightening the security measures, practising the policy plans, having a clear vision to update the policies and regulations, and regularly checking the system infrastructure.

8.4.1 Intrusion detection systems (IDS)

Is detecting unanticipated events that differ significantly from the average behavior, also known as outlier detection. In this context, intruders, also known as hackers, can be categorized as clandestine masqueraders and legitimate or phony application creators [98]. IDS detects intrusive behavior as it helps identify and enable segregation of the compromised devices and warns our system about breaches. As a result, we can reach a point when we roll back harmful changes to the system infrastructure. Both Intrusion and fraud detection systems provide a second layer of defense and play an essential role in securing future computer systems. IDS are classified as active that causes alerts and logs access warning along with instructions to protect the system from changing the configuration when detection occurs, and passive that detects and flags the intrusions. Its primary methods are:

Signature-based IDS This method has a database containing records of attack patterns and relates what happened with what's already stored in the database that can deeply inspect packets for malicious patterns in the packet header or payload. It uses the pattern matching technique to inspect the packets against a frequently updated database of attack signatures to find an attack that matches the records. Nevertheless, its restrictions lie in the new attacks as it only recognizes the existing ones in the database to trigger false negative (FN) rates. The signature-based is either static or dynamic. The static disassembles the code to investigate the structure, and the dynamic execute the code and observe the behavior regarding that all the signature IDS make comparisons between the signature in the records and the current state to trigger a resistance measure when the two states match. In IoE, the fast growth in applications and newly integrated devices make the method of signature-based detection invalid.

Anomaly-based IDS In this technique, we search for vulnerabilities based on rules pre-stated by users by applying artificial intelligence concepts by defining the baseline of a typical working situation and store these attributes in the system at various time intervals. It's a more dynamic intrusion detection system as it can detect unknown attacks by comparing the attributes with the new behaviors. Anomaly-based IDS goes through two main phases, which are training and detection. The cons related to it are generating false positive (FP) attacking reports if it doesn't meet the user set regulations for the system, cost more, and hard to obtain appropriate metrics to govern the

baseline. The higher the false-positive rate for any IDS can cause the real threat to pass unobserved.

Host-based IDS this technique will be installed on the host side to collect data and analyze the traffic when it's going out from our host server. Host-IDS only monitor its installed environment and detects malicious logging behavior. The limitations are that it doesn't provide a whole picture of the system as we only view the host data because it only concerns the host server, limiting the whole network topology and modifying the logs cause the network to fail.

Network-based IDS are strategically placed around the network topology and focus on the network side by monitoring data flow and detecting any irregularity or suspicious activities towards the network side and needs high capabilities of processing collected data together. However, its cons are limited in the variety of the detected attacks as it only concerns detecting the network anomalies that can create a bottleneck in the passing traffic and not immune against protocol-based attacks.

The level and sophistication of today APT attacks for proactive attacks are immense, but when expanded and including deception methods that will deceive the hackers by creating luring baits by simulated environments that grab the attacker attention, which is not part of the hierarchy even if the attacker gets a foothold he will be distracted from the original assets. Honeypot is a mechanism that produces simulated traps complementing the security method as they keep running as typical systems. Their value lies in being attacked and compromised as baits to analyze the hacker behavior that generates signatures by that the security of IDS will be boosted. For example, authorization and communication modules are the most attacked parts in IoV and deploying honeypots can absorb the damage and analyze the attack patterns.

Urias et al. [99] suggested a deception framework that influences virtual and software-defined networking solutions for creating random and flexible deception environments by SDN utilization to adjust the environment as needed dynamically and showing the framework abilities in the case of assailant actions by relating and allowing the networks and end-points performance. Thus, leveraging emerging technologies such as virtualization, cloud, and SDN and adds the ability to study and develop better practices against challenges. In [100] The authors presented an efficient collaborative intrusion detection framework for vehicular internet security consisting of three levels: the RSU, cluster head, and cluster member. In addition to a rule-based decision for trust scheme and the usage of Support Vector Machine (SVM) from machine learning techniques, which detects the anomalies and discussed the detection mechanism and SVM role in extracting, training, and classifying to suit the lightweight

communication scheme. Anomaly recognition can as well be investigated from the perception of extensive data analysis through information theory by considering anomalies as rare events that interrupt the normal system flow that possesses a small probability concealed in the complete information, and these events encompass treasured information. Entropy provides a credible solution for describing the distribution and the status of such events as reducing dimension and the related analyses of events can increase efficiency [101].

8.5 Smart grid mitigation

Drawing a plan for operational risk assessment is crucial to conduct a threat assessment allowing every organization to regulate and protect its assets. That includes evaluating security devices, servers and network devices, and main workstations and sub-stations; the plan must not neglect the site physical security auditing because smart grid cybersecurity incorporates interior and exterior challenges to secure the grid security perimeters from control signals and monitoring the interconnectivity as disturbances can happen unexpectedly. What is necessary to add to the defense scope is leading-edge encryption, physical tamper alerts, secure messaging platform, third-party audit, keys for devices, application-layer protection, strong authentication, and firmware, not just deploying sophisticated gears to create additional entry points for attackers. Network coding guarantees privacy, providing the grid with unlinkability and undetectability to ensure secrecy as confidentiality attacks intend to disclose information. Encryption guards the information that makes the network. The mitigation from packet sniffing can be done by applying security on the gateways that transmit internet protocol packets inside a VPN tunnel so that the regular IP freight will convert to encrypted payload. Transport layer security protocol (TLS) will be used to protect the communications. X.509 digital certificates by the International Telecommunications Union (ITU) can be used for the authentication procedure to manage the identities of different entities inside the grid alongside the users as it generates public and private keys; the key pairs will be used to encrypt and decrypt messages. The detection of malware will be by the aid of anomaly detection tools to guarantee integrity and authentication. The abuse of integrity will lead to safety problems that timestamps can prevent with sequence numbers for the application layer. Regarding the transport layer, an efficient solution is using anomaly detection technique by these two methods we can assure only authorized people are permitted, defense in depth method can add domain-based Message Authentication, Reporting, and Conformance (DMRAC) for email and verification for boundary defense for the information that will be transferred between the

networks while these networks themselves are segmented to keep outsiders.

Synchronized cybercrimes aiming toward the power meter readings are hard to detect because they are unobservable attacks that compromise the grid operation. In [102] Giani and Garcia proposed an effective algorithm to detect these unobservable attacks, which include the usage of a random number of line power meters with injection meters to quickly detects the attacks in the power systems by the use of the phasor measurement units (PMUs) to mitigate and neutralize cyberattacks. Under the assumption that all lines are metered, allowing the meter readings to come out asynchronously in the state estimator, the malicious actions based on the compromised data will be mitigated.

In [103] the authors investigated the integration of electric vehicles in the smart grid from the cybersecurity consequence that causes high voltage instability after cyber attacks by carefully investigating the impact of switching attacks on the infrastructure and the power transmission, shedding light on detection techniques for these circumstances to mitigate the vulnerabilities in the charging station that can destabilize the grid generators. In the central management system (CMS), they installed a Back Propagation Neural Network (BPNN) scheme to detect the switching attacks by enabling the discard or delaying the execution by a controller. The results provided higher accuracy to reach the stability phase. The work in [104] they proposed a mechanism to ensure frequency stability during cyberattacks by providing a switching unit that works as a voltage regulator, which offers complete protection against voltage outbreaks and frequency disturbance. The automatic generation control (AGC) and Automatic Voltage Regulator were analyzed separately and the voltage regulator will calibrate the system to stay within the traditional operation boundaries. The controller will use them as parameters to constantly observe the system and respond with actions. This model is limited to isolated systems.

8.6 IoV mitigation

Converting to using secured routing protocols and the three most common security protocols are: Secure Ad-Hoc on Demand Distance Vector (SAODV), Ariadne, and Secure Remote Password (SRP). SAODV protocol verifies multiple routes for routing the message accompanied by a digital stamp for the key field and one-way hash role to verify the hop count to validate the integrity and authentication of the routing information. Hence, midway nodes can't alter the source node preventing the malicious nodes from sending false data. Ariadne protocol exploits broadcast mechanism relies on one-way hash authentication chain. In this protocol,

the malicious node can't forge false data, and the tesla key computes the MAC involved in the packet. Secure Remote Password (SRP) creates a protected communication that involves the source node and the destination giving them shared keys as password-authenticated key agreement (PAKE), so the client and the user talk independently apart from each other to the server using mathematic calculations, and the server doesn't store the password but treat it as zero-knowledge protocol uses the ignorance of request occurrence which prevents the destination from DOS attacks [105]. Another practice is using flash memory modules that possess secure elements to offer benefits to gain navigation control, control system data and guarantee compliance. In addition, its flexible nature can be adopted in numerous applications that are concerned with cars' cybersecurity and data traffic. Communication of electronic control units (ECU) can be encrypted to protect the maps using flash modules with a security feature in automobile prevention the manipulation and car deviations [106].

Raya [107] presented the revocation protocol of tamper-proof devices (RTPD) using message aggregation and groups as the certificate authority (CA) is accountable for erasing the pair keys and generating a cancelation message. The work is divided into solutions based on asymmetric cryptography and hybrid cryptography, helping to encrypt the vehicle's public key before forwarding it into the other vehicle. Islam [108] proposed a security platform for V2I architecture named CVGuard for cybersecurity that detects and prevents cyberattacks on the environment. CVGuard considered the scalability, flexibility, and upcoming usability with a case study on distributed denial of service (DDoS) attacks to avoid contradiction and overlapping vehicles. Hence, it prevents car accidents because of CVGuard deployment leveraging edge computing, reducing DDoS attacks by 60% at unsignalized intersections. Chen et al [109] introduced an access protocol mechanism in a cryptography game that forces the attacker to answer questions based on the answer. First, the game will initiate a session to solve the cryptographic with the probability that every vehicle must register by birthday paradox. Then, after registering, the simulator will search the queries to check the records when logging by smart cards; else, a random output will be generated with the same probability distribution and adding the record to the list after sending datagrams and this change will be vague for the attacker. Thus, this protocol can ensure authentication for users and data.

8.7 Energy cloud mitigation

Edge or fog computing processes the data in the nodes instead of the data center. This mechanism can add privacy and security attributes to the systems as they have appeared to address cloud computing deficiencies. Additionally, fog

and edge computing are more suited for quicker analysis and project work as using them over the cloud makes the data in the devices hard to tamper with. On the one hand, fog computing mainly handles the time/delay factors or sensitiveness in the applications of the IoE much better [104] (e.g., smart grids and adversarial environment), providing quickness, security, and availability conditions since the work can be done without the internet. On the other hand, edge computing is even more secure than fog as the data will reside at the edge nodes of the architecture. In comparison, the cloud is more suited for long-term data analysis. Vehicle cloud computing has won much interest lately. This paper [110] discussed a detection mechanism based on inter vehicles cloud for detecting potential cyberattacks by malicious vehicles that reserve the benign vehicles privacy. They provided an independent key distribution scheme then a protocol to detect the attacks and tune their frequency in the system based on requirements and showed the effectiveness from the experiment data in real-world which achieved untraceability. The optimization scheme for the green cloud is being adopted to increase the life span of devices and reduce carbon fingerprints in energy infrastructure such as smart meters as the consumption rate and life span are critical factors in the grid sustainability and working endurance. The repeated changing of this infrastructure has economic effects and time factors. Therefore Siddiqui presented a knowledge-based strategy that fetches the frequency lifespan for smart meters based on a customizable graph so the usage status of the grid will place the smart meters in a repository of green-cloud [111].

Edge computing is mature enough to resolve the bottleneck problem of IoE resources by offloading the data concerns in computation and storing storage to end-users which will significantly decrease the response time and traffic flow. Besides, the fears from confidential data leakage will be addressed [112]. Liu et al. [113] proposed a framework for an intelligent mechanism based on an edge-chain that enables access control for IoV between the nodes and RoadSide Units (RSUs) by making the RSU serve as the edge nodes so that the entree will be granted based on access control of risk prediction (RPBAC) and regulate the vehicle behavior. In addition, Generative Adversarial Networks (GANs) convert the risk forecast to a sequence generation. Table 5 shows models and their responses.

9 Future mechanics

9.1 Software-defined networking (SDN)

It started to grow more popular to deal with distributed systems where we have nodes considered to be the brains

Table 5 Mentioned models response and gained benefits

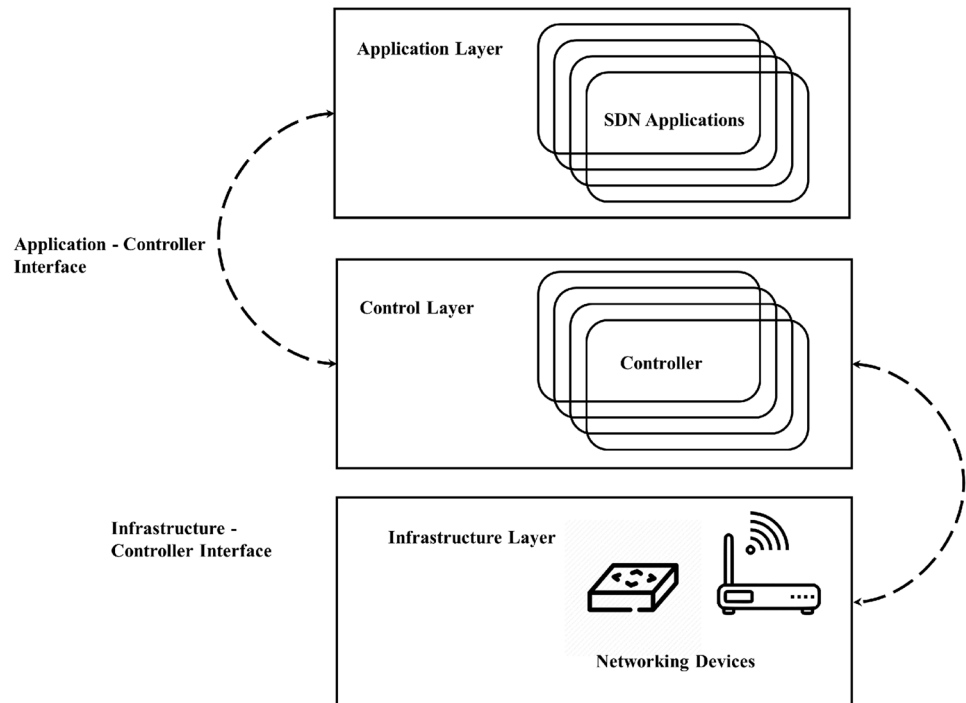
The technique	Gained benefit
Enumeration	Identifying resources and discovering vulnerabilities
Multi-Factor Authentication (MFA)	Integrity and authentication
Time-Division- Multiplexing (TDM)	Availability and reliability
Physical Security	Integrity
Rule-based Anomaly Detection System (RADS) [91]	Achieved 90–95% accuracy (F1 score) with a low false positive rate of 0–3% in real-time
Mechanism in RPL Networks [92]	Standard RPL recorded 40% packet loss rate and the parent fail over scenario has a packet loss rate of 0 to 2% providing integrity and availability.
Datagram Transport Layer security (dtls) protocol [93]	Authentication and end-to-end security, very small headers, average 15% less energy is used to send and receive
Concept of Capability (DCapBAC) [94]	Integrity and legitimacy of each device
Fully Homomorphic Encryption (FHE) [97]	Less computation time, deal with various messages to gives higher privacy and protection
Lightweight Intrusion Detection Framework for Vehicular Networks (AECFV) [100]	AECFV exhibited a high detection rate, close to 99%.
Two-Stage Protection Method [103]	restricted the magnitude of the oscillations between 0.18% to 1.2% in different scenarios
Automatic Voltage Regulator [104]	maintained the terminal voltage within permissible limit in comparision to simple AVR
CVGuard [108]	Reduce DDOS by 60%
knowledge-based strategy [111]	72% enhancement in lifespan and 21% minimization in energy consumption

of the SDN. Its principle lies in programming and management by separating the control logic inside the network by the function of forwarding data to obtain better performance and monitoring [114]. It's changing the map of how networks deployment and operation works significantly inside centralized intelligence by controlling and forcing the policy across all the network aspects and stricter the authentication and system accessibility while maintaining and managing the system efficiently by pushing the controls from the hardware side to remains exclusively software-based [115]. Figure 8 shows the SDN reference model.

Since SDN customizability can collect more data from monitoring to facilitate the network intelligence, which can address the cloud demands for flexibility and agility as the networking infrastructure solution can provide these requirements. It even provides innovation to make more green innovations such as energy-aware infrastructure and energy-aware routing to be adopted. It is worth pointing out that SDN provides adaptable northbound application program interfaces (API) that enable users to develop more secure applications or use third-party services to monitor the network behaviour and control it [116]. The combination between SDN and cloud is built upon network virtualization as the critical key for this structure. After all, it will govern the resources sharing to ease the management, enhance the QoS and harden the cloud by introducing automation in its structure to eliminate its threats because

the intersection networking is flexible and quick for reconfiguration in accordance with the software instructions. [117] an approach for SDN with edge computing software-defined edge computing (SDEC) was advocated which decouples upper-level applications from the basic physical edge computing by defining them as software, making the correspondence between the edge devices and the virtual models increasing the edge heterogeneity by cyber-physical mapping. SDEC main roles are to descending the management of physical resources via virtualization and ascending various edge resources by programmable functions to serve as an operating system for the edge physical hardware that can share, reprocess, and reconfigure. Smart grids will also rely on the networking between the data acquisition devices, sub-grids, smart meters, and supervisory control to transfer a massive amount of data. Therefore, SDN will change the interfaces and the awareness of the grid as the capabilities of easing the management by connecting various and large numbers of nodes, packet classification features for cross-domain exchange, and virtualization to isolate entities to exercise separate logical control based on requests [118].

SDN's ability to mitigate cyber-attacks will come from the control plane abstractions and network function virtualization from a security perspective. They will provide a global topology for networking that monitors the real status and simplifies the packet forwarding and control the programs and packets for communications. There lie big

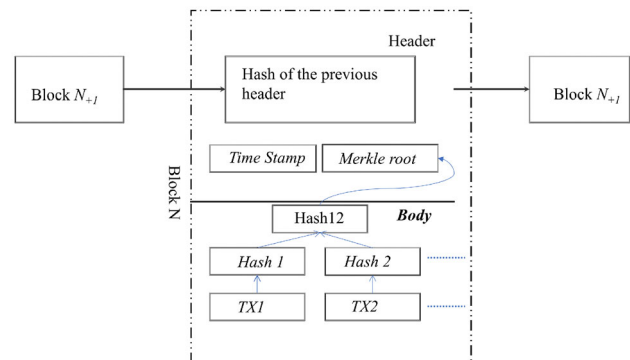
Fig. 8 SDN reference model

benefits from the centralized structure of the controller and the virtualization will empower SDN by initiating fast routing response or instruct devices to enable encryption during the forwarding phase. On the other side, the designers should consider the communication and computation overhead very carefully as the efficacy of processing the packets can cause a single-point failure [119]. It is worth noting [120] proposed an SDN controller in vehicular edge networks with an attack tolerance mechanism based on deep reinforcement learning to comprehend the control role and convenient management for the complex inter-vehicular architecture. Further, illustrating the policies to protect the vehicles.

9.2 Block-chain

Blockchain is a distributed ledger that includes a collection of blocks chained together that records different data records. In it, we can generate a new block and attach them to the chain. It can play a dynamic protagonist as a democratized platform with no need for a third party, and excluding third parties will make the confidence grow up in users [121]. The need for central trusted authority is gone and replaced with multiple entities that can create and store a chain of blocks as every entity can verify that the order and data are still intact. Figure 9 describe the blockchain structure.

Another prerequisite to change blocks that the hash of that block will alter differently if any of its content changed, and each block has the previous block's hash to be

**Fig. 9** Blockchain structure

considered. Hence, it's almost impossible to change the chain maliciously. In this cryptography technique, the final node can only know the key. The blockchain practicalities on Peer-to-Peer will be groundbreaking for IoE Optimization as it suits the distributed renewable resources and smart grid deployment, making IoE more secure, fast, transparent. When updating a node, the update will spread through the chain. Block-chain highlighted features that it's [122]: open source application which can encourage many individuals to create even more secure schemes, fault tolerance, transparent, immutable inside the block for outsiders and anonymity. Since the first appearance as a method to secure Bitcoin and its payment, we can employ it for IoE payment and this immutable secured payment mechanism to protect the transactions. It takes advantage of the concept that each block inside the chain can be

adjusted to meet a specific problem's specific requirement. It can operate by many algorithms that can be personalized for modern decentralized architecture, such as Delegated Proof-of-Stake and Proof-of-Authority [123]. However, now it's tough to be integrated directly due to the insufficient computational resources. However, it's applicable for energy cloud based on fog or cloud computing. As the network in blockchain consist of 4 nodes where every node has different duties to support the services, mining, and routing due to the peer-to-peer architecture (P2P). This integration can provide scalability, reliability, and autonomy. Taking the security phase forward by combining SDN control and blockchain on edge computing, producing a novel model reducing computational burden [124]. IEEE introduced a framework for Internet of Things (IoT) based on Blockchain Data Management, this framework identifying the building block in the blockchain during IoT data life-cycle, demonstrating block interactions and data acquisition [98].

We can classify the blockchain by attribute; if the chain allows new blocks to join, engage, and access contents with the old blocks, it's known as permissionless. Unlike permission chains due to their structure, only the predefined and authorized personal are permitted. Immutability and auditability features in blockchains add another dimension of synchronization between the network as the only way to modify the ledger are by changing the majority of the blocks to be malicious ones. In the blockchain context, this technology exploits cryptographic techniques making all the records kept safe by asymmetric cryptography, which works well for IoE smart grids. The decentralized nature addresses some known problems that designers always considered in terms of resiliency and fault tolerance because there is no single point of failure. Finally, the secure and the script deployment, also known as smart contracts, are a bunch of codes written and can execute automatically after satisfying pre-stated measures without human intervention.

For IoV [125] the authors implemented identity-based access controllers named MECUs (Mother Electronic Control Unit) so all the other electronic control unit forward their data for the mother unit which has the ability to add blocks to the network chain after the performance of security checks, then the verified MECU double-check the CIA triad in each ECU. Making the blockchain work only in the mother ECU to save storing resources.

The author [13, 126] describes a blockchain structure for secure vehicle-to-vehicle communication; in this structure, all the existing vehicles keep broadcasting their place via beacon messages to surface the targeted messages containing information such as the driving statuses, then a location certificate is issued for identity proof. The proposed model posses two-segment known as a broadcasting

model and a mining model. On the one hand, the broadcasting model is done by vehicles broadcasting messages among themselves during an event and the peer vehicle authenticates the event message. On the other hand, the mining model is concerned with authenticating message reliability to ensure integrity. Wang et al. presented a mutual authentication for key agreements based on blockchains for edge computing for smart grids to allow this protocol to support anonymity and keys supervision without sophisticated cryptographic that ensures integrity and authentication. This protocol highlight is presenting an efficient key update and overturning mechanism and reducing the computation cost [127].

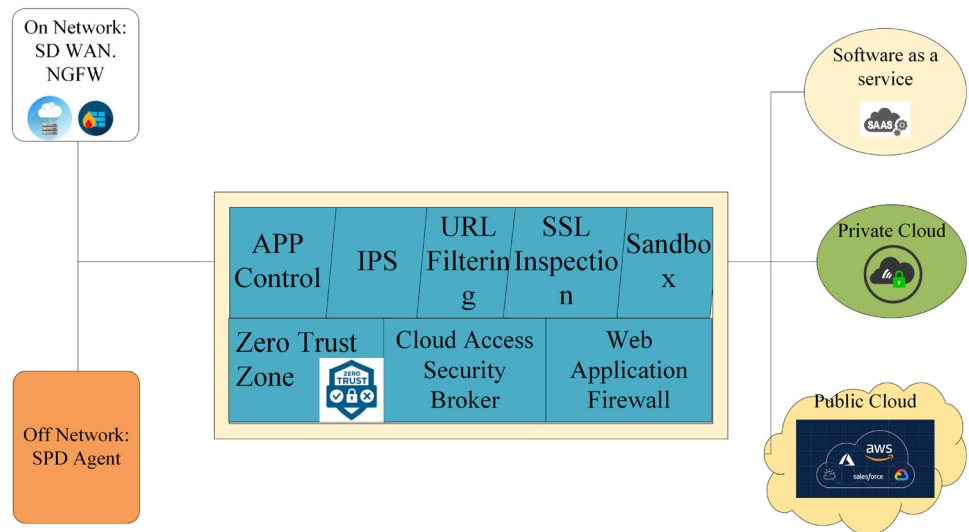
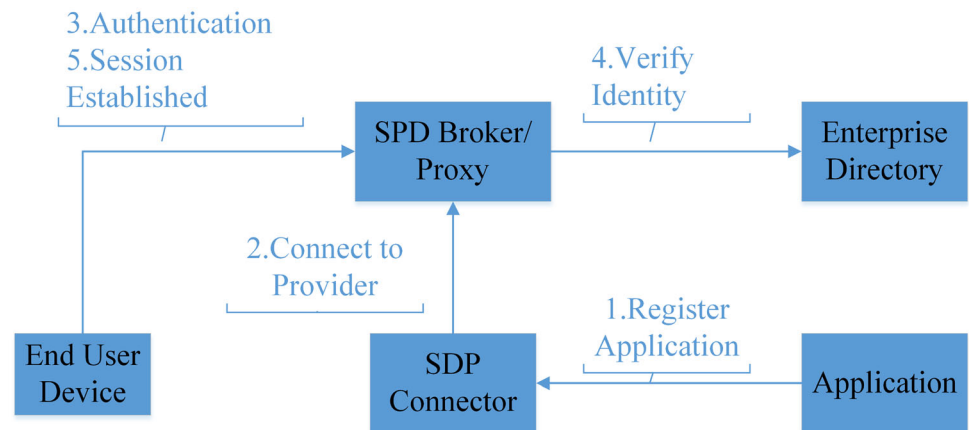
9.3 Secure access service edge (SASE)

SASE will be the new architecture that blends the VPN and WAN in the network, using the previously discussed SDN, cloud security, and WAN. Resulting in a new cloud defined well by software [128] that prevents data loss and offers domain-name-service. SASE architecture is shown in Figure 10. This new cloud infrastructure will deliver optimized performance, exceptional flexibility, and more security in all levels with capabilities such as Zero Trust Network access given the least-privileged access according to policies as an edge network, software-defined wide-area networking (SD-WAN), filtering contents, cloud access security broker, and optimized routing.

SASE is the convergence of different technologies such as next generation firewalls is stressing the least privileged principles that can change the current security face of hiding behind the traditional firewalls by treating every asset as a stand-alone network edge – resulting in isolation and invisibility from outsiders.

The Zero Trust architecture shown in Fig. 11, is embedded inside and will keep all the undesirable access from reaching the network as it sees each device with its own identity for protection, assuming no implicit trust granting and just focusing on the security pilers (security, confidentiality, and integrity). Segmenting the network will only permit the authorized person to access a specific application rather than accessing the whole network with a large portion of remote users. In this case, SASE will assume that every device is hostile as most data breaches happen because of blind trust.

The security broker works as a warranty for more security between users and the cloud; the cloud governance the service, and there are embedded firewalls as a service which is a firewall for the cloud as an extra layer of protection within some cloud applications needs internet connection producing an application-aware security policy that is enforced on a large scale to gain granular control. SD-WAN will describe the management of the WAN as a

Fig. 10 Delivered SASE architecture**Fig. 11** Zero Trust Network (ZTN) access conceptual model

software as a service scheme as it differs from the traditional WAN by preventing the traffic accesses explosion vulnerability that can bring the following benefits: extensions to public clouds, provides higher availability, aware routing to match the need of locations, policy compliance, protection and secure broadband, and separating the network from the applications. Converting from traditional firewalls to next generation firewalls (NGFWs) will provide much efficient security as they have a far better solid and intelligent structure because of the traditional firewalls recent incidents as they are placed at the network borders and, once bypassed, fails to protect. What attracts that their ability to simplify tasks and policy controlling for users aside from that the possession of remarkable features that differs from the traditional firewalls as they contain all the standard features and extra features such as virtual private network (VPN). The traditional ones keep logging of the sessions when active, but when the sessions end, they see themselves inbounded to protect. At the same time, they don't keep logging records for unsuccessful sessions. For

comparison, NGFWs adds geolocation attributes to associate IPs with the actual physical locations, embedded intrusion detection systems (IDS), and sandbox environments. When downloading a file, it places it in a virtual machine environment and executes it after checking its safety, then passes it down to the destination. Also, it has a web application checker that protects the application layer, which prevents SQL injection and cross-site scripting attacks. Inside the NGFWs, heuristic analysis functions will be embedded to search for commands that usually should not be found in an application indicating a virus mechanism or a worm pattern. If the instructions behave like a virus in the emulated environment, then possibly it's a virus, by using this multi-criteria analysis by given available data to find unknown data since the viruses that don't have a signature need sophisticated techniques and heuristic analysis provides real-time analysis to block it at the gateways.

9.4 Machine learning techniques

The development of computer-based research is achieving a great breakthrough and keeps attracting a lot of research interest; besides the enormous growth in the number of connected devices that demand fast resource allocations, ML can serve those purposes by anticipating and reserving resources in emergencies or as deployment as a protective shield for our cybersecurity solutions. ML can be sub-branches to three, namely, supervised learning, unsupervised learning, and semi-supervised learning. Supervised learning uses labeled input that will produce a labeled output and unsupervised learning has an unknown targeted class that lacks some information concerning data sorting. Semi-supervised is the hybrid combination of both types. ML techniques have two operational phases: training and testing. In training, calculations and measurements are performed for the data over a while and acquire or learn its behavior. In testing, the tested cases are classified as normal or intrusive based on the acquired learning. Machine learning efficiency with intrusion detection was demonstrated based on the decision tree classifier dealing with malware detection; the author showed how decision trees displayed the best accuracy of 99.96% and capabilities to discover illicit behavior [129]. More investigation regarding ML solutions for cybersecurity by these authors [130, 131] displayed ML role in intrusion detection and capabilities will serve the detection of botnets and Domain Generation Algorithms (DGA). The works investigated the critical analysis of supervised, semi-supervised and unsupervised classification by comparing three models and their usage with cybersecurity as botnets are chains of connected bots used to steal data, send spam, and probe for weaknesses. The person in charge of the chain is called a bot-herder. On the other hand, DGA is an algorithm used by cybercriminals to allow their domain to connect and surpass blocklisting by generating domains. From all the above, we can understand that ML roles can be in classification and spam filtering or forensic analysis or generating inputs to test programs structure for weaknesses [61]. Especially the forensic analysis portion, which has been neglected in the past and allows criminals to escape. Also, deep belief networks (DBN) have now been used to mitigate cybercrime by making every layer communicating with the previous layer and the layer after it, excluding the node's communications [132]. ML and AI can reduce the cost of predictions in networking because traditional network nowadays uses a predefined set of monitoring probes, contrary to normal algorithms ML produce its complexity report depending on the data quality and size. Its adaptive nature will balance the quid pro quo between performance improvement in many terms such as QoS, congestion

control, and computational cost. Utilizing SDN can empower ML methods to offer the picture-perfect platform to realize adaptive probing. We can have cyber threats prediction by ML through implementing some tools such as Anaconda, Spyder, and Python [21]. Furthermore [133] put a novel ML framework based on SDN and Network Function Virtualization (NFV), which can adapt to many scenarios with anomaly-based IDS; the anomaly detection rate was 99.71%. Deep generative modules like generative adversarial network (GAN) have the capability to produce intrusion signals to simulate the anomaly to train the system which can be a vital technique, just like creating new algorithms to reach robust detection [17].

9.4.1 Isolation forest

An algorithm based on the decision tree method that possesses remarkable capabilities in identifying anomalies uses partitions to split the data and determine the feature, then it takes the randomly selected data and split them into maxima and minima. The algorithm comes with an anomaly score to evaluate situations that separate anomalies from normal points. After creating a decision tree, it proceeds to isolate the observations which overcome the normal isolation difficulty and reduce the computation as any anomalous behaviour is a sudden dip or spike in the metric. Thus, this unsupervised technique overcomes the need for feedback and previous information in supervised learning. Additionally, it can be seen as the most acceptable detection algorithm that deals with big data for the time being because it possesses the best-in-class performance and works better with data outliers based on the receiver operating characteristic curve (ROC), which indicates the performance of the model classification abilities [134]. Other detection algorithms start by building a profile containing the typical instance and flagging anomalies that do not fit and deals with low dimensional data, while the isolation forest isolates the anomalies based on each tree computation score and makes an average across all trees to deal efficiently with high dimensional data. The working principle that this recursive algorithm will keep dividing and subdividing the data (x) to make it as homogenous as possible according to an attribute Y and split value P at each subsample till one of three scenarios firstly, reaching a high limit by the tree, secondly, the observation gets isolated on the sides of the outer node. Lastly, the data x reach the same values as all the attribute. Eventually, observations will get isolated per capital. The path length for observation will be calculated by $h(x)$, which is the collections of isolation trees (the path) from the root node until the termination phase in the outer node resulting in three scenarios:

- The observation score is close to one, the length of the pathway is small, and detection of an anomaly occurs then data points need isolation.
- The observation score is less than 0.5, the length of the pathway is considerable, and no detection of an anomaly occurred; then this data point is normal.
- The observation score is roughly close to 0.5; then these samples don't contain anomaly at all.

The algorithm law is: $s(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$ where $E(h(x))$ is the average, (s) is the anomaly score, n is the number of outer nodes. Thus, this formula will generate a monotonic relationship between the score and the average, and the internal nodes number is $n - 1$. The isolation forest doesn't rely on distance or density, making it fast, inexpensive for computation, and utilizes less memory [135].

10 Future directions and challenges

Some of the future mechanics to address IoE security are still in their infancy due to the lack of IoE-relevant experiential data and IoT-specific attack signs as data protection is of paramount significance in IoE realms. Here we talk about them:

For blockchain, the execution is still challenging for academics, operators, and practitioners due to limitations in adoption, though several works are being done, as an example the throughput issue, which is defined as a statical form for the transaction processing time per seconds, is throwing its shadows in the ability to make transactions by blockchains because of its implementation in cryptocurrency and the relation with the type of blocks whether it's permissioned or permission-less. In contrast, private and permissioned chains possess higher throughput than public and permissionless chains, albeit these private chains can't provide full decentralization notions. They are frequently deployed under a central collaboration with decentralized chains with lower throughput. Another challenge is the consensus mechanisms, as consensus algorithms have high costs in computation, reducing the energy consumption, increasing the efficiency, and making it multi-purposes instead of the single purpose application currently trending. Another test is the smart contracts; those contracts are a form of programming codes that embedded inside the chains and having such established in advance entities in the chain can cause a vulnerable state to be used by malicious attacks as the code themselves potentially contains vulnerabilities, bugs, and trusting issues if a third party wrote it. In the blockchain, if a hacker investigated the vulnerabilities and analyzed the available information about users actions and energy profiles, he/she can establish a pattern for the user actions or energy consumption,

and other records like restricted information can get traced and leaked even perhaps revealing users identities since transactions are in a ledger.

Future research direction in ML with cybersecurity can be expanded on approaches such as transferred learning by making models for tasks like intrusion detection. In this technique, The models are reused or become the foundation for newer models as an entry point for other tasks by retraining (tuning the model) to improve the performance because it deals proficiently with models that needed to be trained with lesser data in new environments. Switching from the traditional ML models that deal in insolation without passing the gained knowledge to leverage newer models can do related tasks in contextual knowledge format mimicking humans. Hence, increasing productivity, reducing time, reducing cost, and reduces the risks, for instance, a nation grid failure will make us able to predict the political impact. In Machine Learning, its algorithm usually manages the inputs that why layers identification, parameters, topologies are complex [136]. The possibilities of an aggressor who uses viruses based on ML that can evolve are growing—at least at present—it's limited because they train Neural Networks (NNs) to mimic legitimate communications. Therefore, the designing methodology needs to be shifted to solve the cybercrimes and security concerns instead of concentration on a single problem on topics like detection. Combining deep learning and encryption technologies with ML is necessary to have a bigger picture centered on security. It will enable us to identify a larger area of attack vectors and evaluate the efficiency of the obtained cybersecurity by its performance metrics. Further, the training datasets need to be open-source datasets and more significant in size than the typically used ones that are way smaller than those that can simulate the actual cybersecurity environments. By the same, the need to address the challenge related to the algorithms as they required hardware that needs to deliver high-end performance and the cost concern with such hardware.

Cloud computing future direction should pay special attention to the connectivity problem since the cloud is based totally on internet connectivity. However, the recent attacks for cloud structure also changed the direction to targets the technologies cloud inside such as disk partitions, CPU, GPU, caches and shared fundamentals because it's designed for compartmentalization and cloud segment this hardware from the network side. In addition, the platforms or the controlling interfaces need to be more secure and enable various interactions schemes to offer adequate security that assures users privacy without neglecting the role of third parties to delivers reliable energy service. Further, improving the regulations in terms of services by government legislations and organization legislations and

have a response mechanism for the cases that fail to comply can also help mitigate espionage. The process of cloud migration to relocate the resources across it can create the risk of channels malicious attacks [137]. Fog is capable of smoothing the adoption of various access control models like access-list or attributed access control in a centralized structure fog will authorize access and convey data between authorized parties; also in the distributed architecture, it will grant tokens to allow authorized individuals.

SDN adaptation should consider the implementations as the need to separate the onboard routing from the switches hindering its widespread utilization [138]. Also, SDN relies totally on high-level policy to replace the traditional low-level network configurations so the concerns about policies contradictions and conflicts from various scopes. Nevertheless, at the same time, it's revolving around the TCP/IP architecture that why it doesn't support the QoS requirements greatly and a compromised SDN controller will be able to change the topology configuration. It also supposes that the forwarding devices are already trusted devices that create a gap in proactive cybersecurity as there is no solution in SDN case to predict attacks in advance before happening [139]. SDN controller compromisation can lead to an invisible darknet as a backdoor for communications for accessing other entities such as field devices to plant rootkit in the kernel, making the attacker able to dodge detection. These SDN rootkits can paralyze the monitoring and control functions and this could be mitigated by the strategic distribution of out-of-band detectors at the network [140].

On one side, the internet of vehicles should look further into the mobility issue as many used sensors were designed to be static and the automobile moving nature makes it interact with different neighbours and generate reports that sometimes include first-time interaction, which should be reliable to generate decisive decisions in a short period of time that has impacts since the services will be transferred from infrastructure servers or nodes to other servers. The prime security challenges are authentication and privacy. The vehicular edge cloud sees these nodes as entry points for the cloud services that will produce confusion by hackers once attacked and compromise the reliability, which demands stronger encryption for the information. Additionally, the tremendous data volume produced by the sensors is quite a challenging task containing plenty of different parameters that require intelligent algorithms in designing and secure operative communication between different schemes such as V2X and V2V, considering this as an open paradigm for researchers and industry to share information to develop the IoV for more reliable integrity and authentication.

On the other side, smart grids future direction should focus on advanced key management schemes promptly to upgrades the keys and include valuable feedbacks from the energy sector sides. The metering communication schemes between entities need more self-aware routing protocols to find the optimal path to cope with the demand management and demand response. A deeper investigation about the RES heterogeneity decreases the control complexity for a better secure by design mindset that can produce reliable delivery and network throughput. More awareness campaigns for the customer sides since they lack enough knowledge and the involved human factor will help to gain benefits from the deployed countermeasures and fast response to isolate the compromised entities. Additionally, realizing that many commonly used industrial protocols such ICCP and Modbus revolve around connectivity only so they can be used to initiate attacks unless more secure protocols that consider security first used [141]. Privacy-preserving scheme concepts need to be explored more so that consumers can decide what they want to keep confidential and maintaining privacy, even if it's a cloud-based smart grid. The Field Area Network (FAN) adds the ability to connects the energy managements systems to utilities and substations without neglecting the isolation, optimization, supply and demands criteria, and distributed resources making it an ideal candidate for future smart grid power delivery schemes. Smart grids protection will increase significantly after securing the communication and providing each device with a unique identity and multi-factor authentications alongside encryption [135].

11 Conclusions

The billions of things that can be connected by IoE mark it as one of the most significant development by humankind as our world today is energy-based to the fullest. However, any step in the direction of a new revolution can get obstructed by some obstacles and here we are facing the security which is hindering IoE growth from shifting towards a complete energy evolution. Hazards will also evolve alongside the technological progress because it's rich with sensitive information or work within sensitive entities and any mischievous attack can hold a global tragedy. We presented a survey for IoE perceptions, challenges, and threats focusing on security core considering the main components: the Smart Grids, Internet of Vehicle and Cloud, and their relationship with IoE. Combining different mechanisms such as Cloud, Software Defined Networking and Block-Chain to address the flaws in traditional structures as it's more concerned with service delivery than security and protection. We presented several attacks alongside mitigation tactics from academic and

industry practices and good strategies to contain the damage and to ensure more security with practical solutions that we must start to include in our structure to reserve the data and privacy rights and to help in designing better security frames in terms of impacts and effects. We believe that this classification is significant to have a clear bigger picture of security issues occurrence to facilitate more design based security for IoE seek.

Acknowledgements This work is supported by the National Natural Science Foundation of China (61973193, U1964207, 61527809), and the Innovative Research group of National Nature Science Foundation of China (61821004)

Author contributions Prof. YL, Supervise the findings of this work and Final approval of the version to be published, MMM role was the design of the work with prof. YL and expanding the concept, Drafting the article and making a mind map, DLM role was classifying papers to help the work and writing with MMM. All authors reviewed the results and approved the final version of the manuscript.

Funding This work is supported by the National Natural Science Foundation of China (61973193, U1964207, 61527809), and the Innovative Research group of National Nature Science Foundation of China (61821004).

Data availability No Data set have been taken which was the basis of another work and all the citation and supplied information referenced where needed.

Declarations

Conflict of interest The Authors declares that there is no conflict of interest, have no other financial relationships with any organizations that might have an interest in the submitted work; have no other relationships or activities that could appear to have influenced the submitted work. The manuscript will not be submitted elsewhere: The manuscript has not been submitted or published anywhere.

Informed consent The Authors give their consent for the publication of the article with all the details within the text to be published in Cluster Computer Journal and may be available in both print and on the internet, and will be available to a broader audience through marketing channels and other third parties. Therefore, anyone can read the material published in the Journal.

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